



(12) **United States Patent**
Kessler et al.

(10) **Patent No.:** **US 12,386,181 B2**
(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **WEARABLE PUPIL-FORMING DISPLAY APPARATUS**

(71) Applicant: **Raytrx, LLC**, Tulsa, OK (US)

(72) Inventors: **David Kessler**, Rochester, NY (US);
Michael H. Freeman, Tulsa, OK (US)

(73) Assignee: **RAYTRX, LLC**, Tulsa, OK (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/406,856**

(22) Filed: **Jan. 8, 2024**

(65) **Prior Publication Data**
US 2024/0310631 A1 Sep. 19, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/138,240, filed on Dec. 30, 2020, now Pat. No. 11,906,736.

(Continued)

(51) **Int. Cl.**
G02B 27/01 (2006.01)
G02B 13/00 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC **G02B 27/0172** (2013.01); **G02B 13/0025** (2013.01); **G02B 13/0065** (2013.01); **G02B 13/0095** (2013.01); **G02B 27/0176** (2013.01); **G02B 27/0179** (2013.01); **G02B 27/283** (2013.01); **G02B 27/286** (2013.01); **G02B 30/35** (2020.01); **G02B 30/36** (2020.01);

G06F 3/013 (2013.01); **G06T 7/50** (2017.01); **G09G 3/007** (2013.01); **G02B 2027/0123** (2013.01); **G02B 2027/0134** (2013.01); **G02B 2027/0138** (2013.01); **G02B 2027/0159** (2013.01); **G02B 2027/0187** (2013.01); **G09G 2340/0407** (2013.01); **H04N 23/90** (2023.01)

(58) **Field of Classification Search**

CPC **G02B 27/0172**; **G02B 13/0025**; **G02B 13/0065**; **G02B 13/0095**; **G02B 27/0179**; **G02B 27/283**; **G02B 27/286**; **G02B 30/35**; **G02B 30/36**; **G02B 2027/0123**; **G02B 2027/0134**; **G02B 2027/0187**
USPC **353/7**; **359/630**; **349/11**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,357,775 A * 12/1967 Appeldorn **G02B 15/145113**
3,614,456 A 10/1971 Hamisch
(Continued)

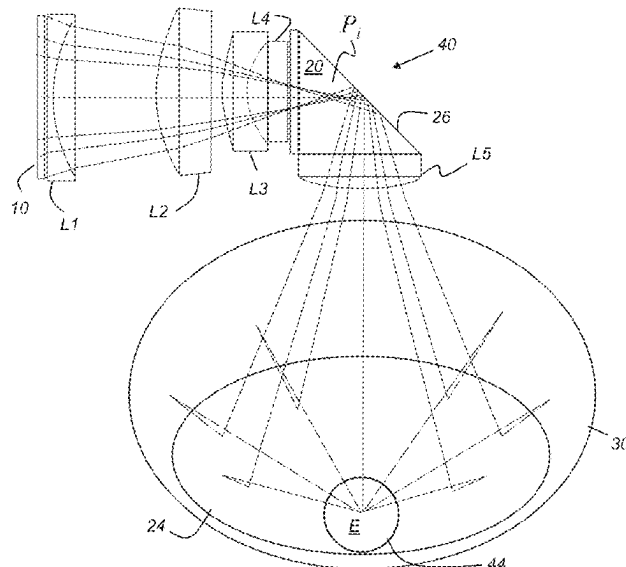
Primary Examiner — Audrey Y Chang

(74) *Attorney, Agent, or Firm* — Howard & Howard
Attorneys PLLC

(57) **ABSTRACT**

A wearable display apparatus is described herein. The wearable display apparatus includes an image generator that forms a 2D image, a partially transmissive mirror having a curved focal surface and a partially transmissive curved reflective surface, a beam splitter, and an optical image relay positioned between the image generator and the beam splitter. The optical image relay includes a right-angle prism, a first plano-aspheric lens, a doublet positioned between the first plano-aspheric lens and the image generator, a meniscus singlet lens positioned between the doublet and the image generator, and a concave-plano field lens.

20 Claims, 21 Drawing Sheets



Related U.S. Application Data			
(60)	Provisional application No. 63/060,343, filed on Aug. 3, 2020.	6,310,736 B1	10/2001 Togino
		8,089,568 B1 *	1/2012 Brown G02B 27/0172 349/96
(51)	Int. Cl. <i>G02B 27/28</i> (2006.01) <i>G02B 30/35</i> (2020.01) <i>G02B 30/36</i> (2020.01) <i>G06F 3/01</i> (2006.01) <i>G06T 7/50</i> (2017.01) <i>G09G 3/00</i> (2006.01) <i>H04N 23/90</i> (2023.01)	10,732,407 B1	8/2020 Wood et al.
		11,112,611 B1	9/2021 Kessler et al.
		11,181,747 B1	11/2021 Kessler et al.
		11,906,736 B1 *	2/2024 Kessler G02B 27/283
		2003/0081317 A1	5/2003 Katsumata et al.
		2006/0055886 A1 *	3/2006 Cobb G03B 21/005 353/7
		2010/0109980 A1	5/2010 Tohara et al.
		2010/0214635 A1	8/2010 Sasaki et al.
		2013/0127980 A1	5/2013 Haddick et al.
		2015/0070773 A1	3/2015 Wang et al.
		2015/0147684 A1	5/2015 Heidt
		2018/0003977 A1	1/2018 Mir et al.
(56)	References Cited U.S. PATENT DOCUMENTS	2018/0039052 A1	2/2018 Khan et al.
		2018/0284441 A1	10/2018 Cobb
	4,311,366 A * 1/1982 Clark G02B 13/24 359/733	2019/0018236 A1	1/2019 Perreault et al.
		2019/0219818 A1	7/2019 Mattinson
		2019/0227319 A1	7/2019 Trail et al.
		2019/0339528 A1	11/2019 Freeman et al.
	5,663,833 A 9/1997 Nanba et al.		
	6,084,715 A 7/2000 Aoki et al.		

* cited by examiner

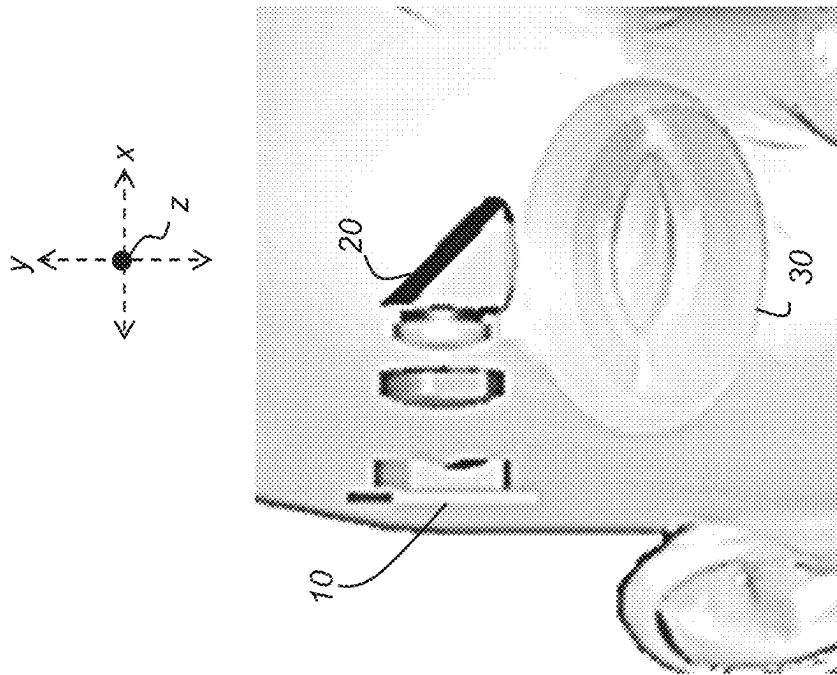


FIG. 1A

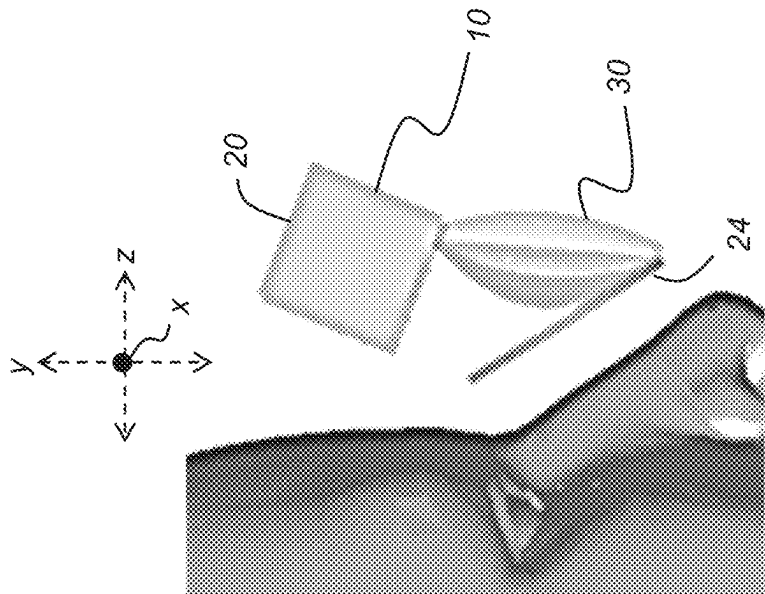


FIG. 1B

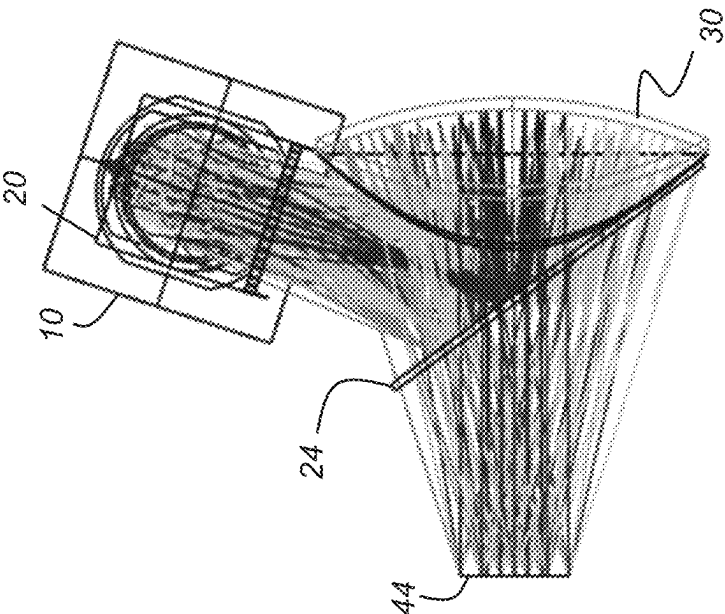


FIG. 2A

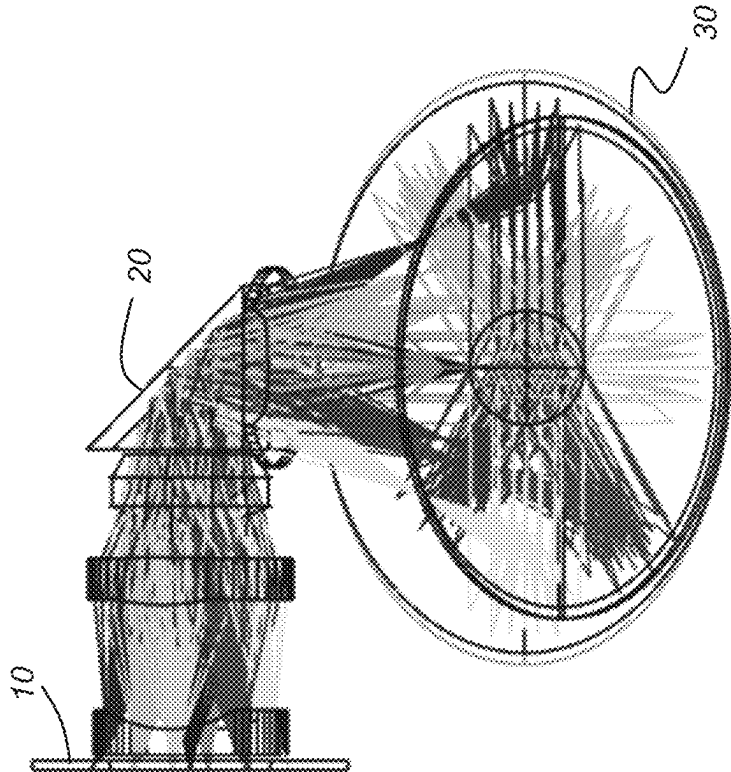


FIG. 2B

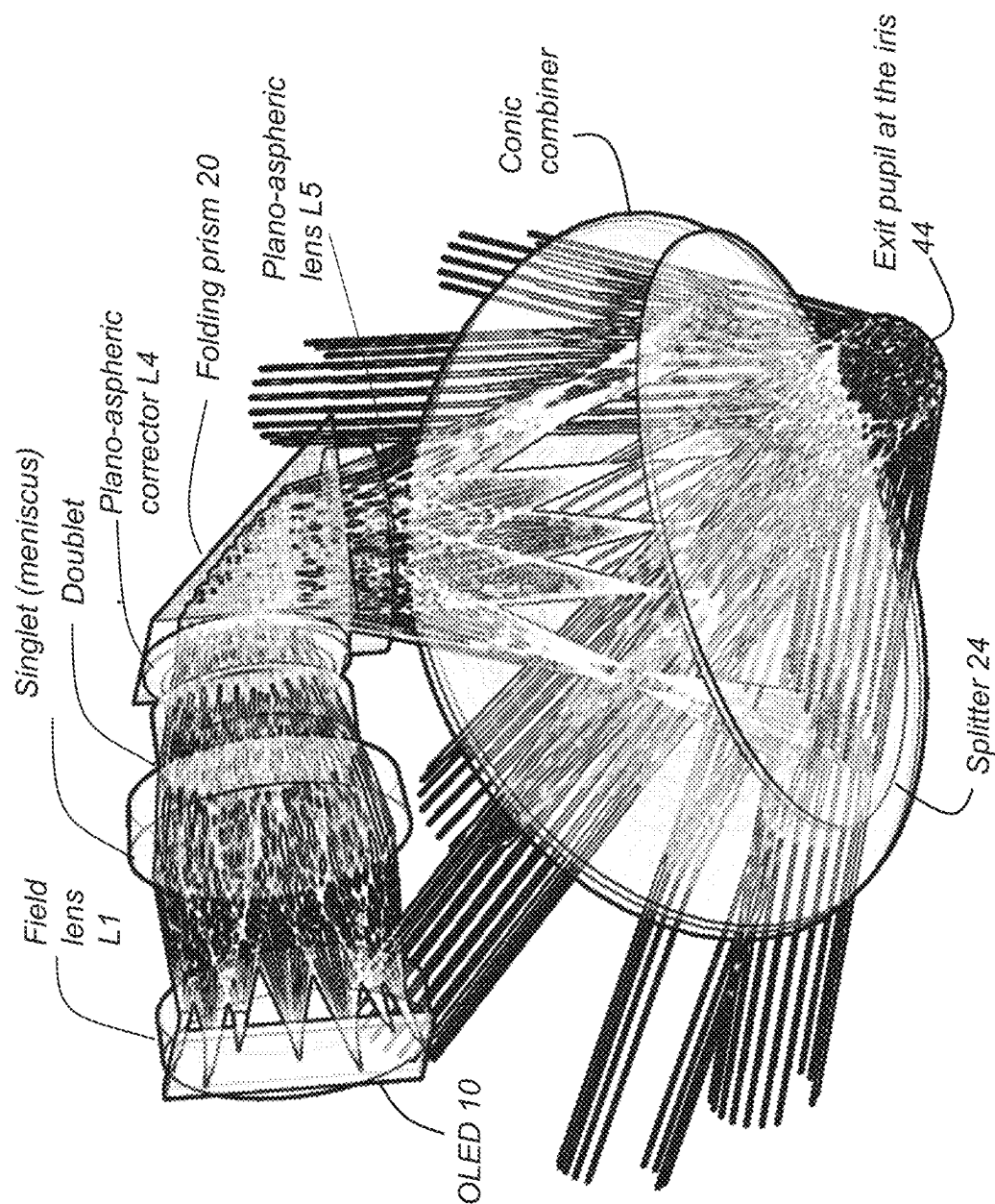


FIG. 3A

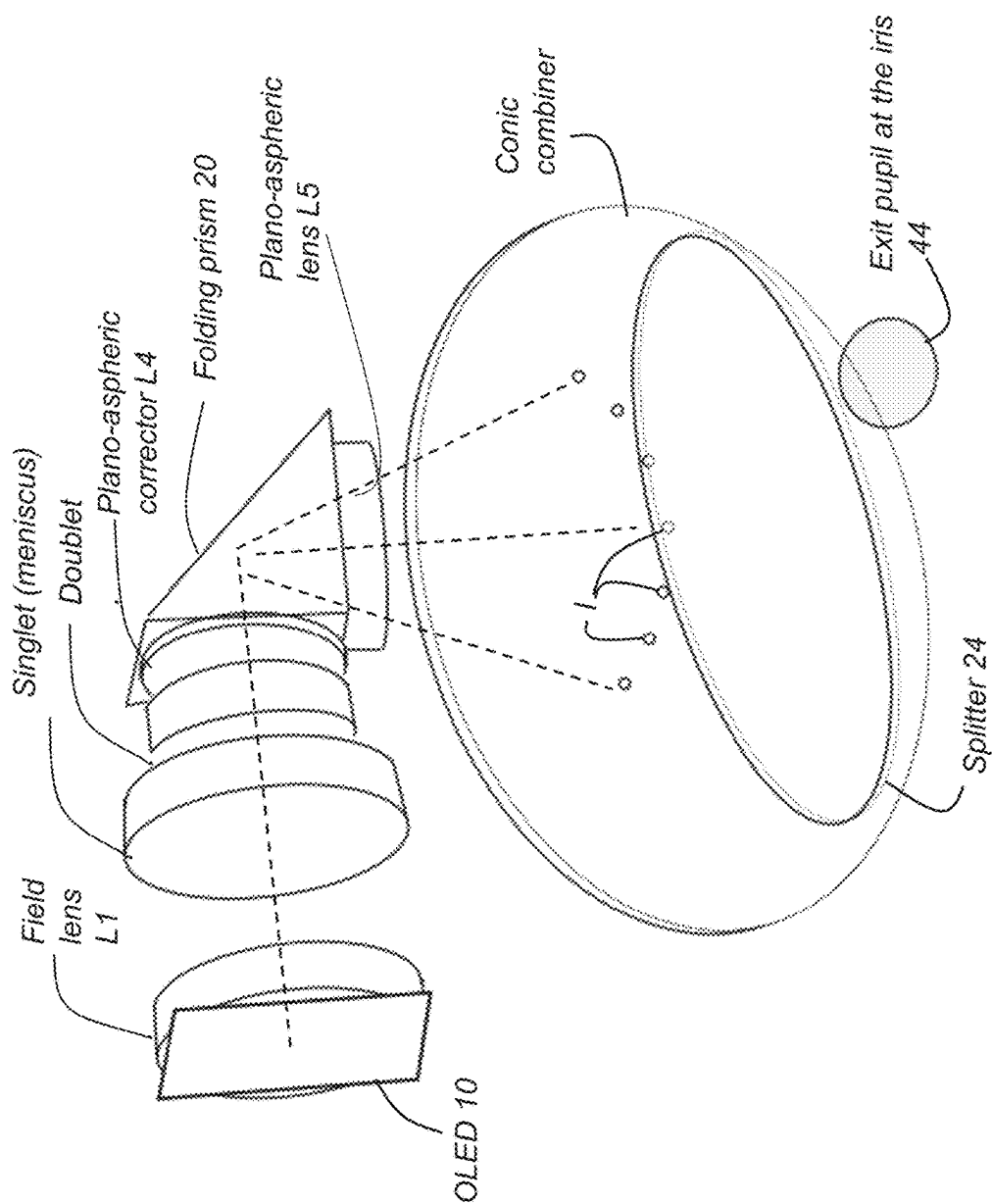


FIG. 3B

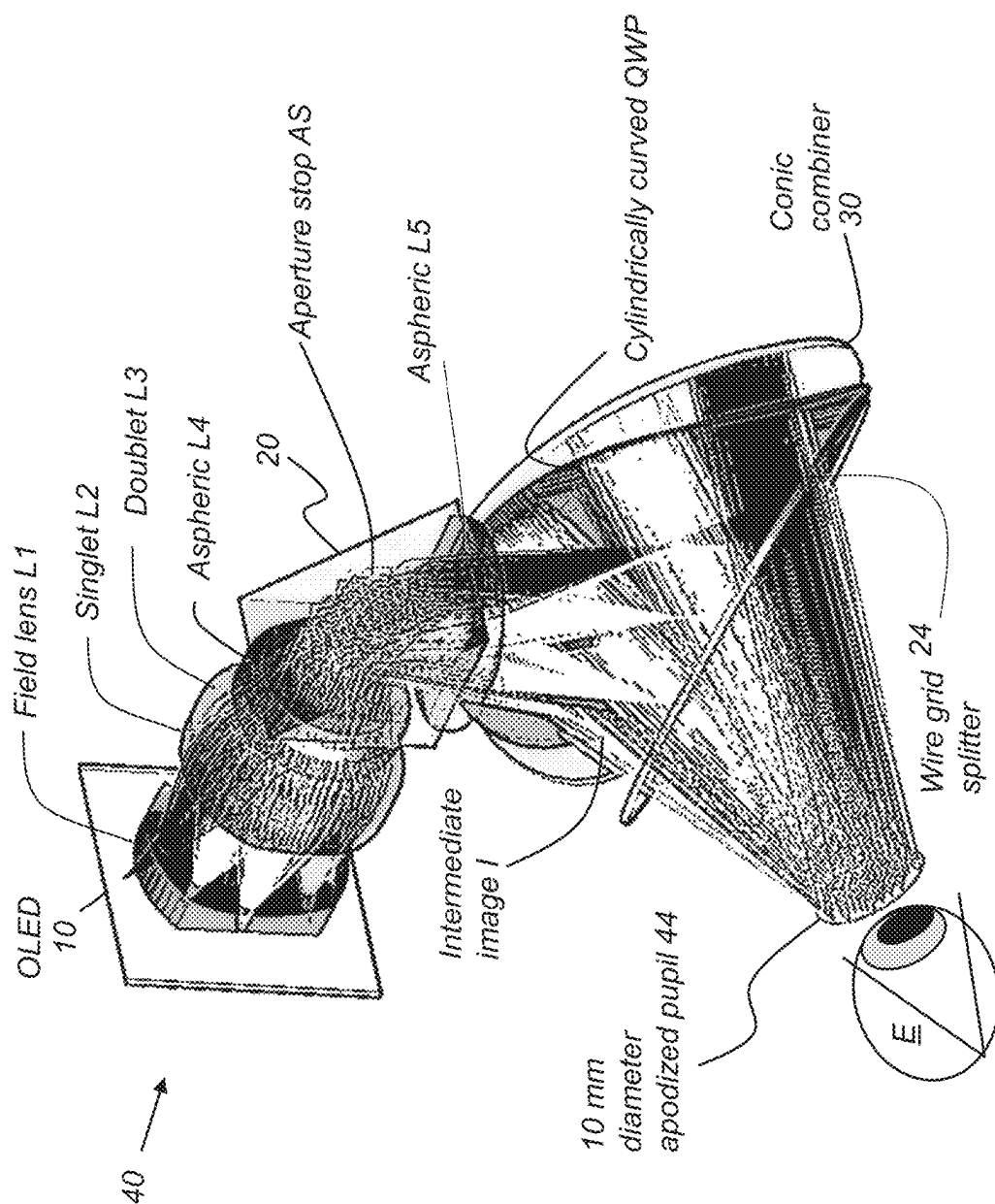


FIG. 4

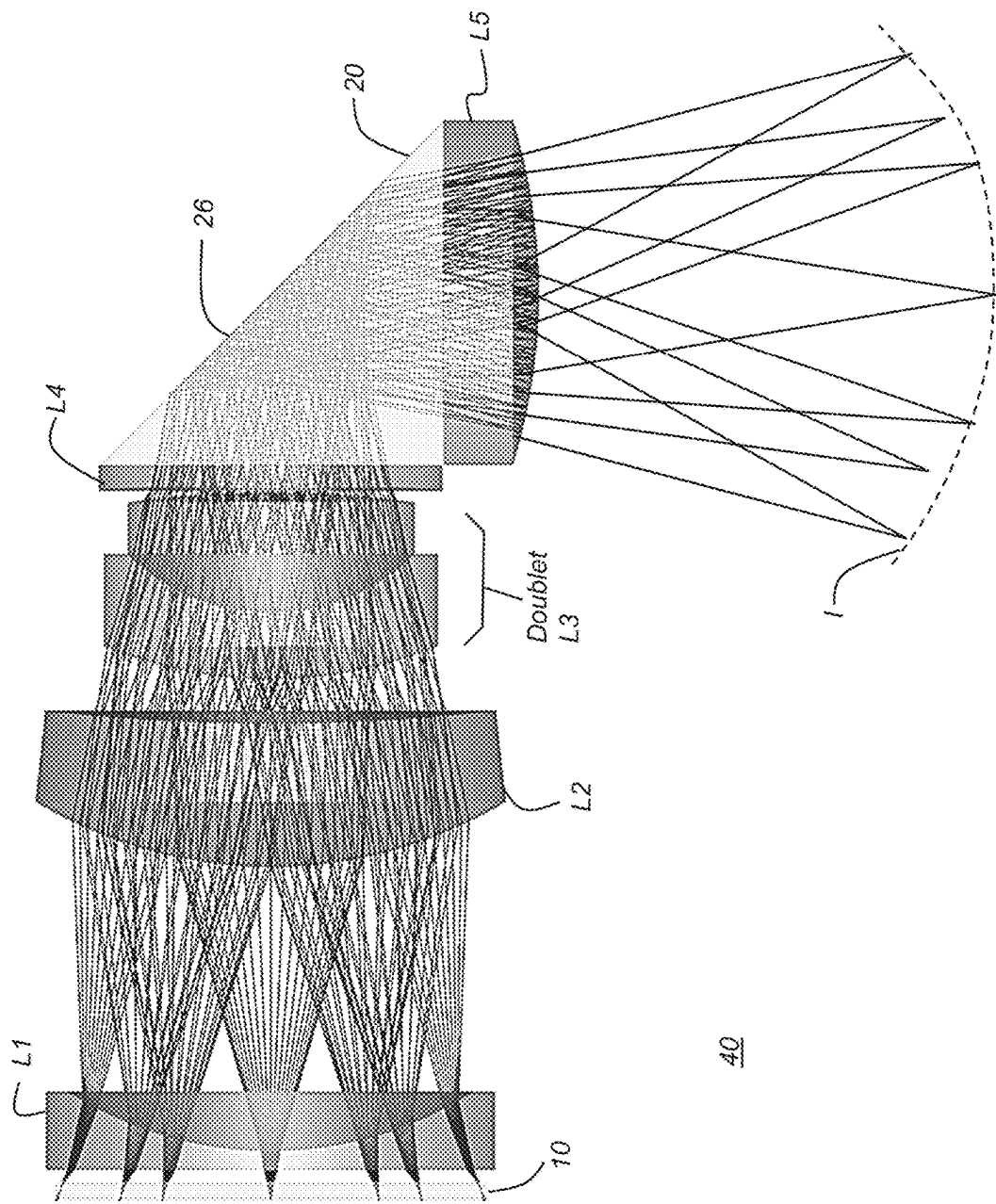


FIG. 5

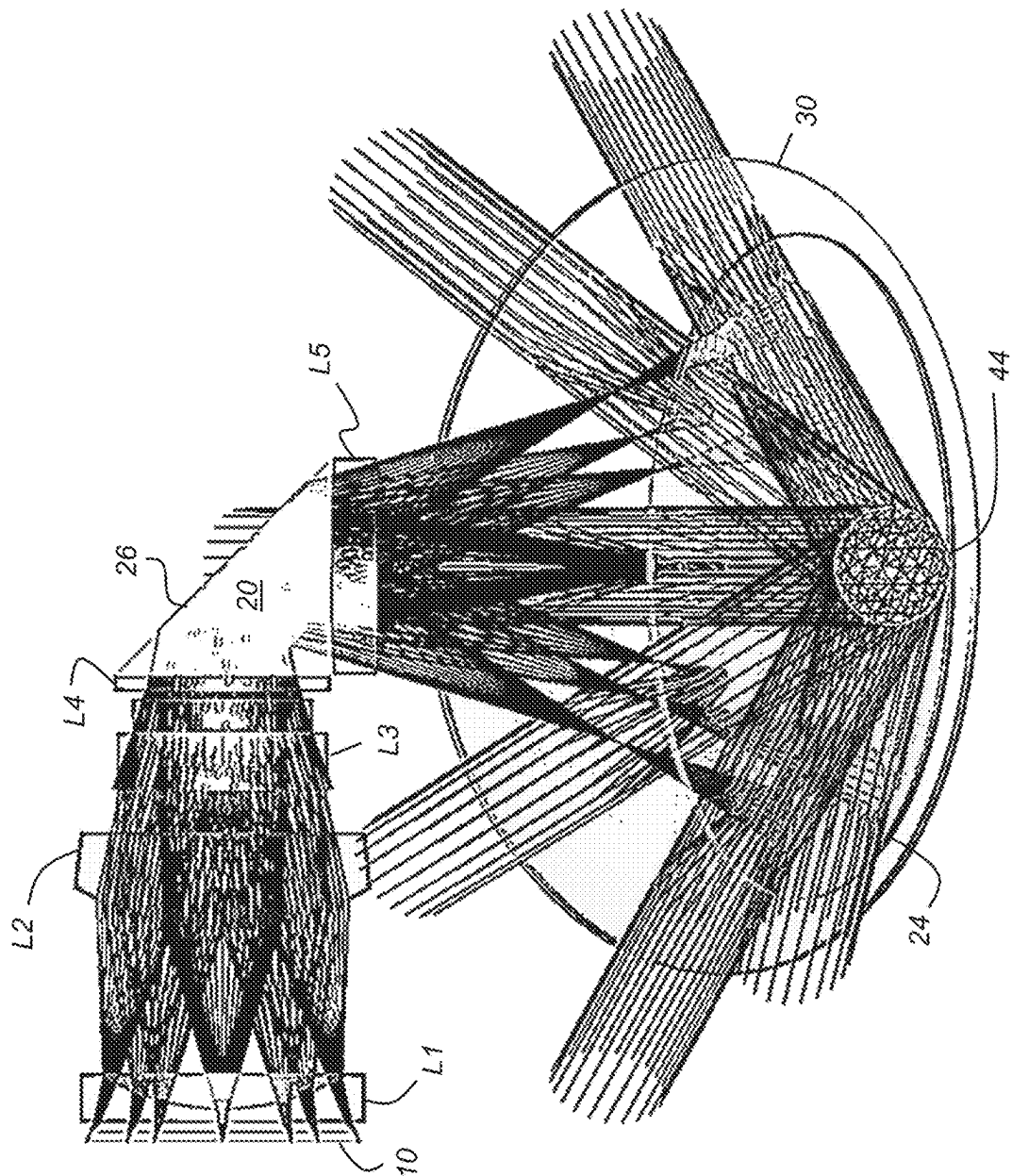


FIG. 6A

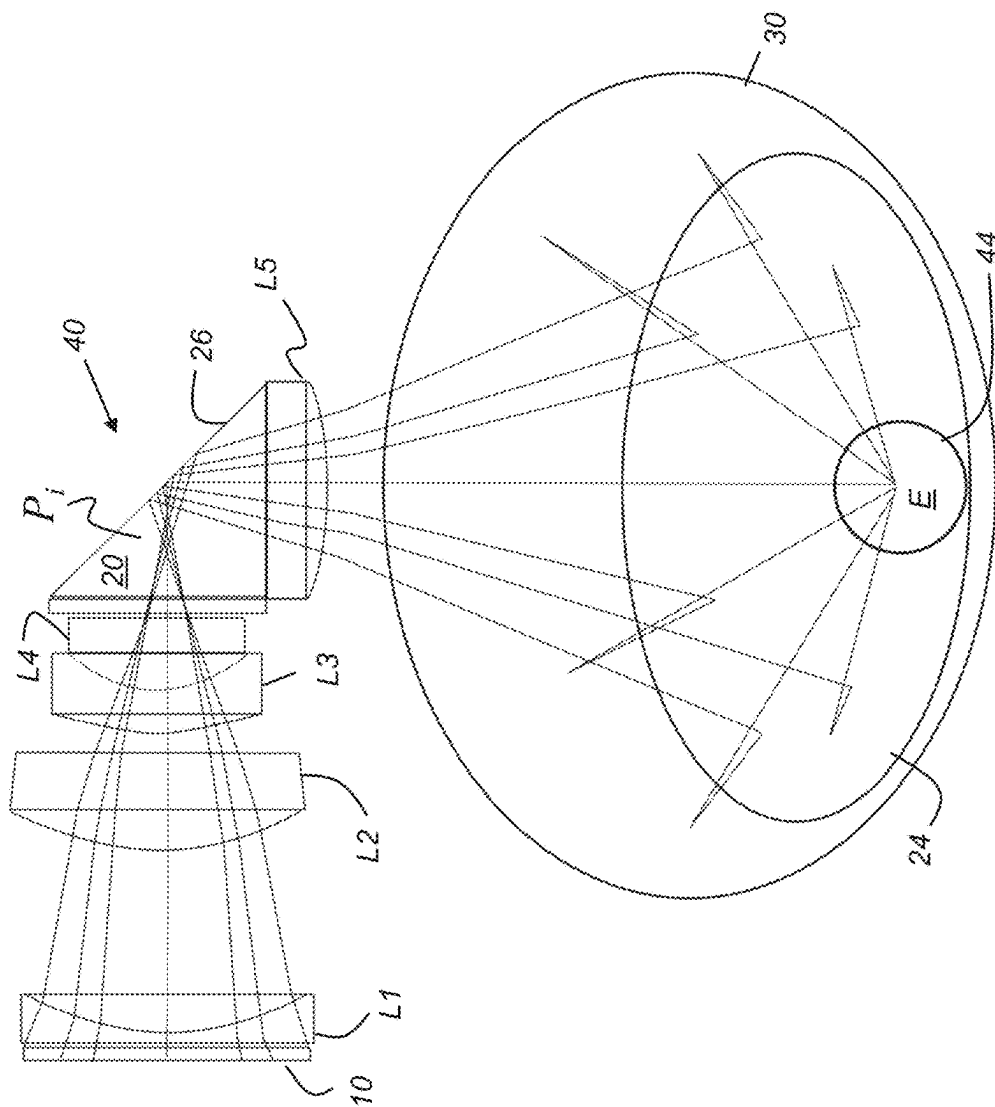


FIG. 6B

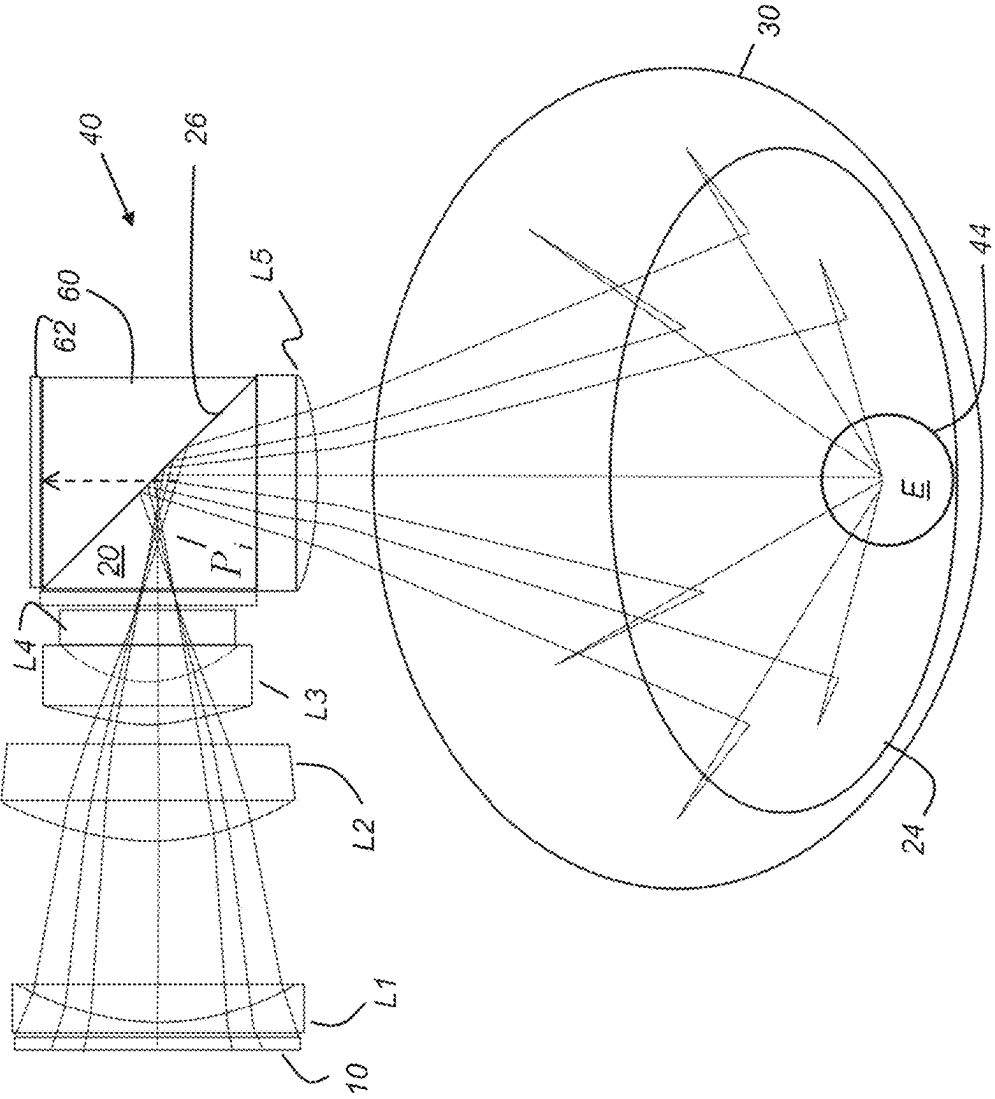


FIG. 6C

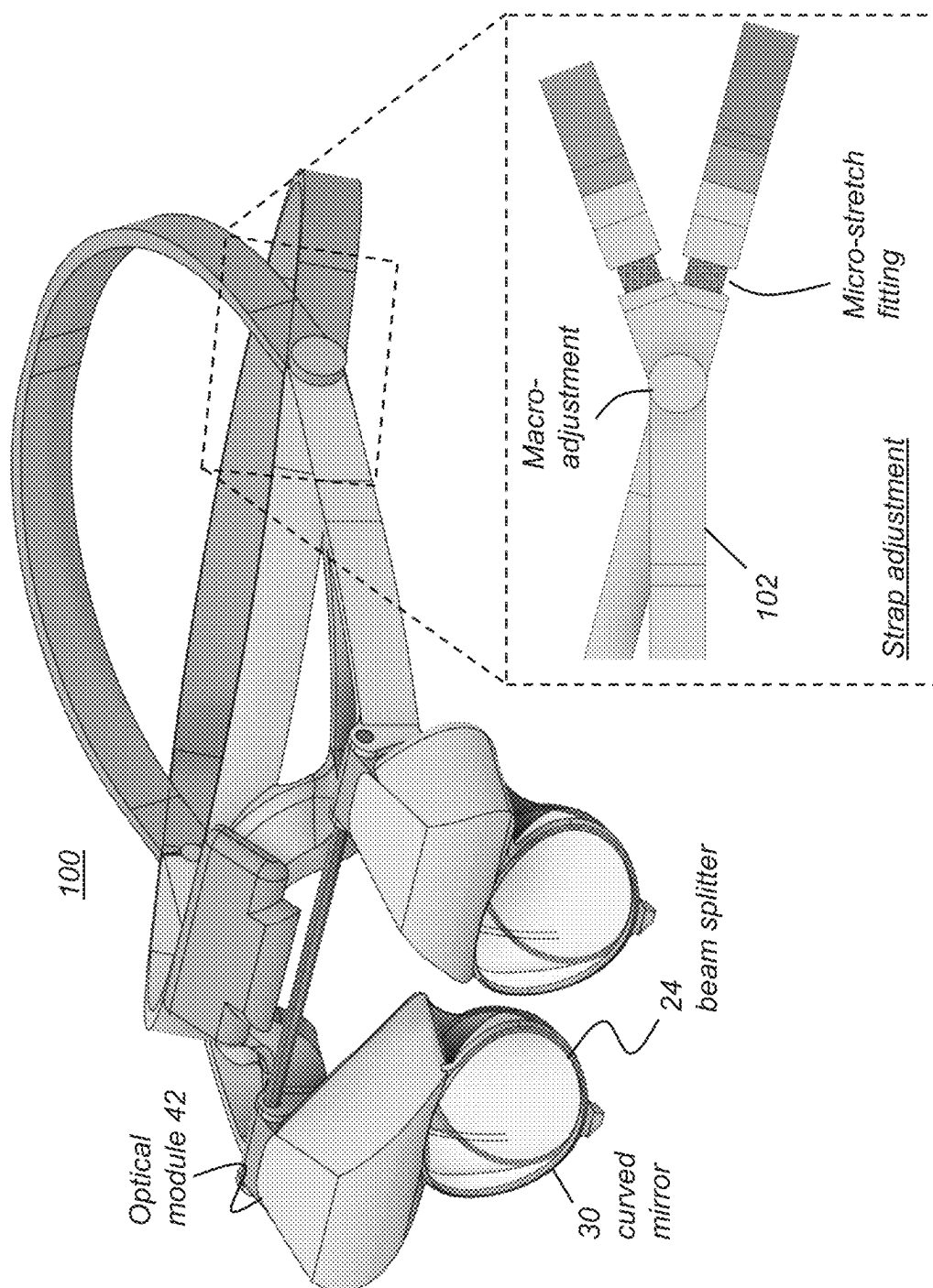


FIG. 7A

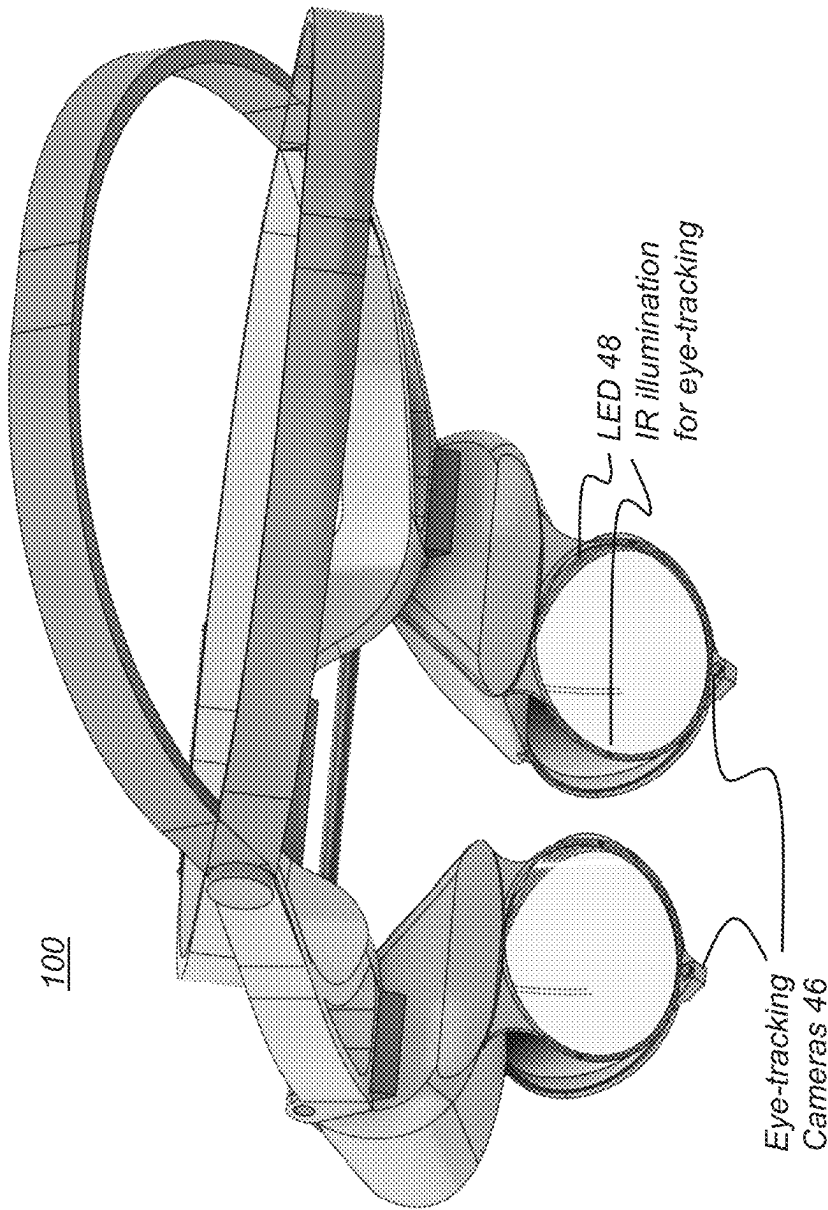


FIG. 7B

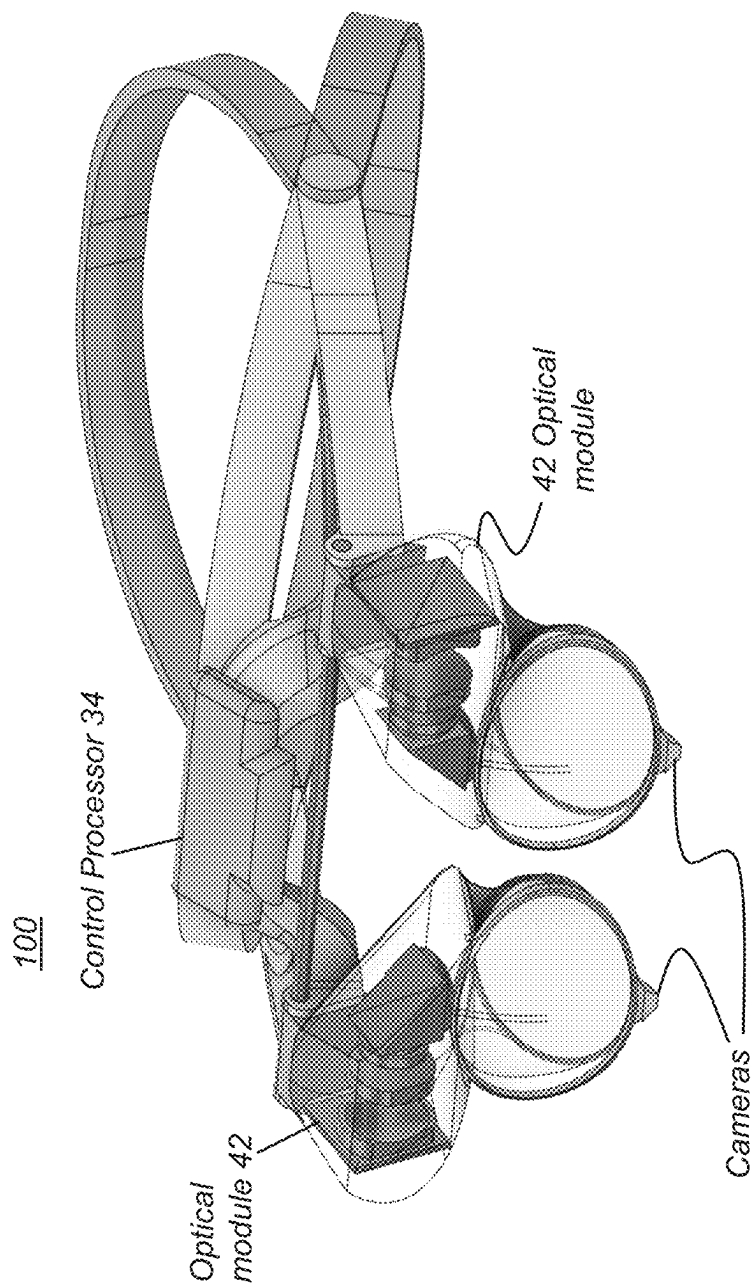


FIG. 7C

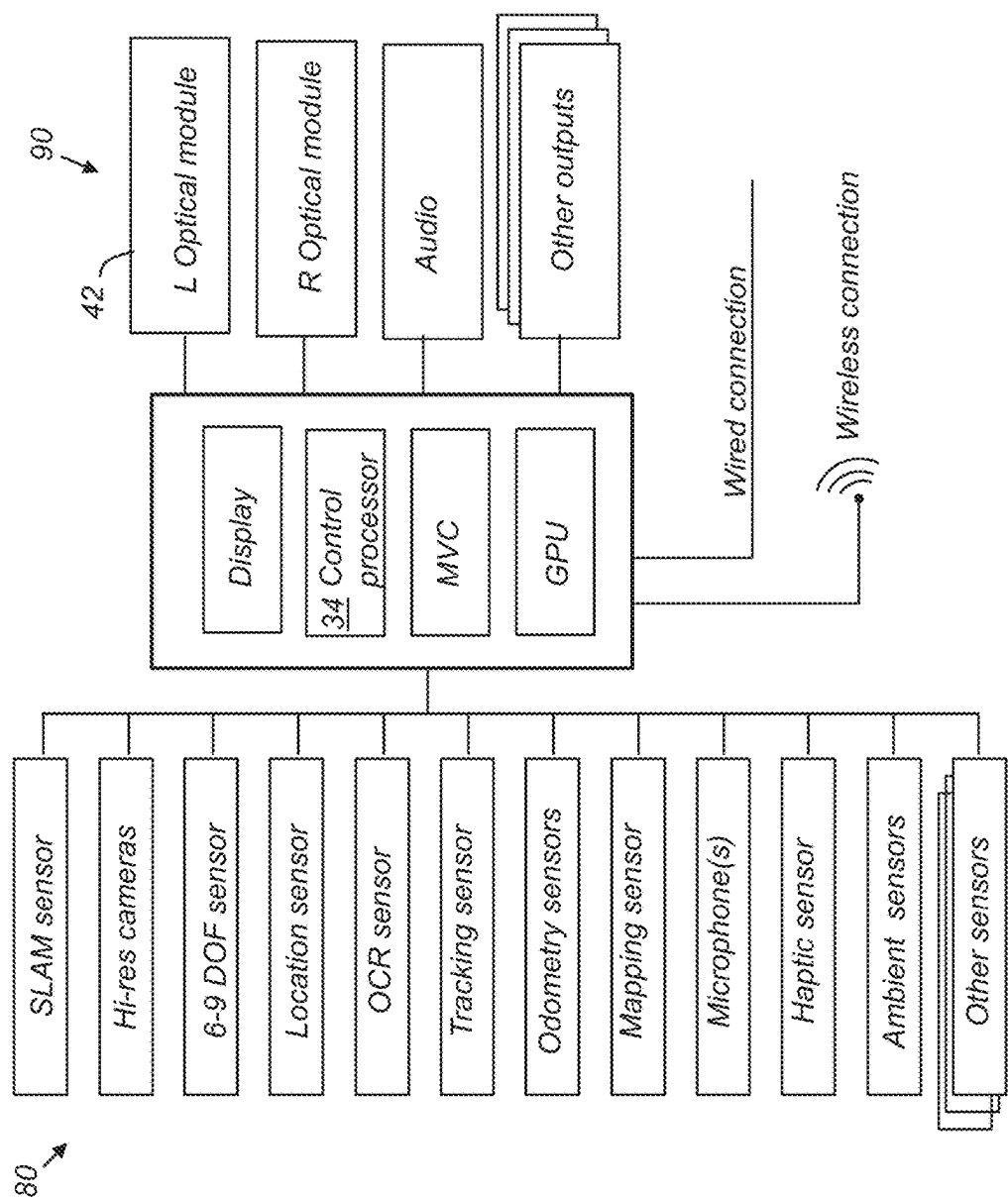


FIG. 8

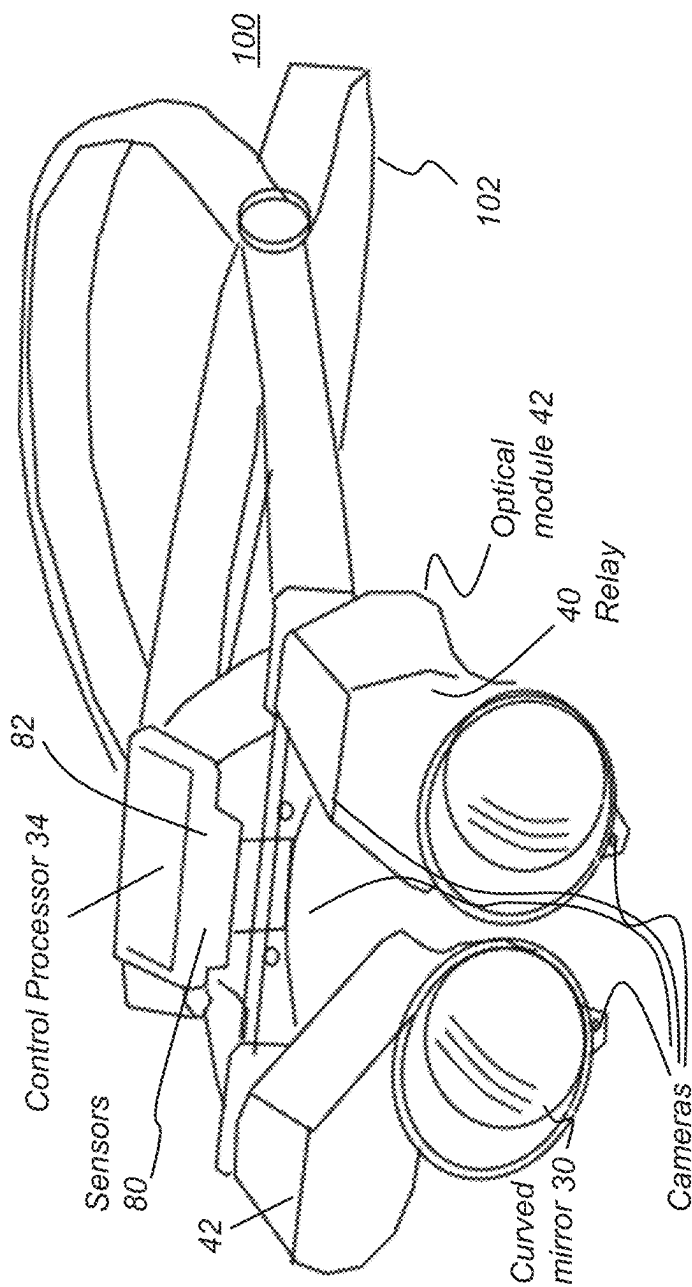


FIG. 9

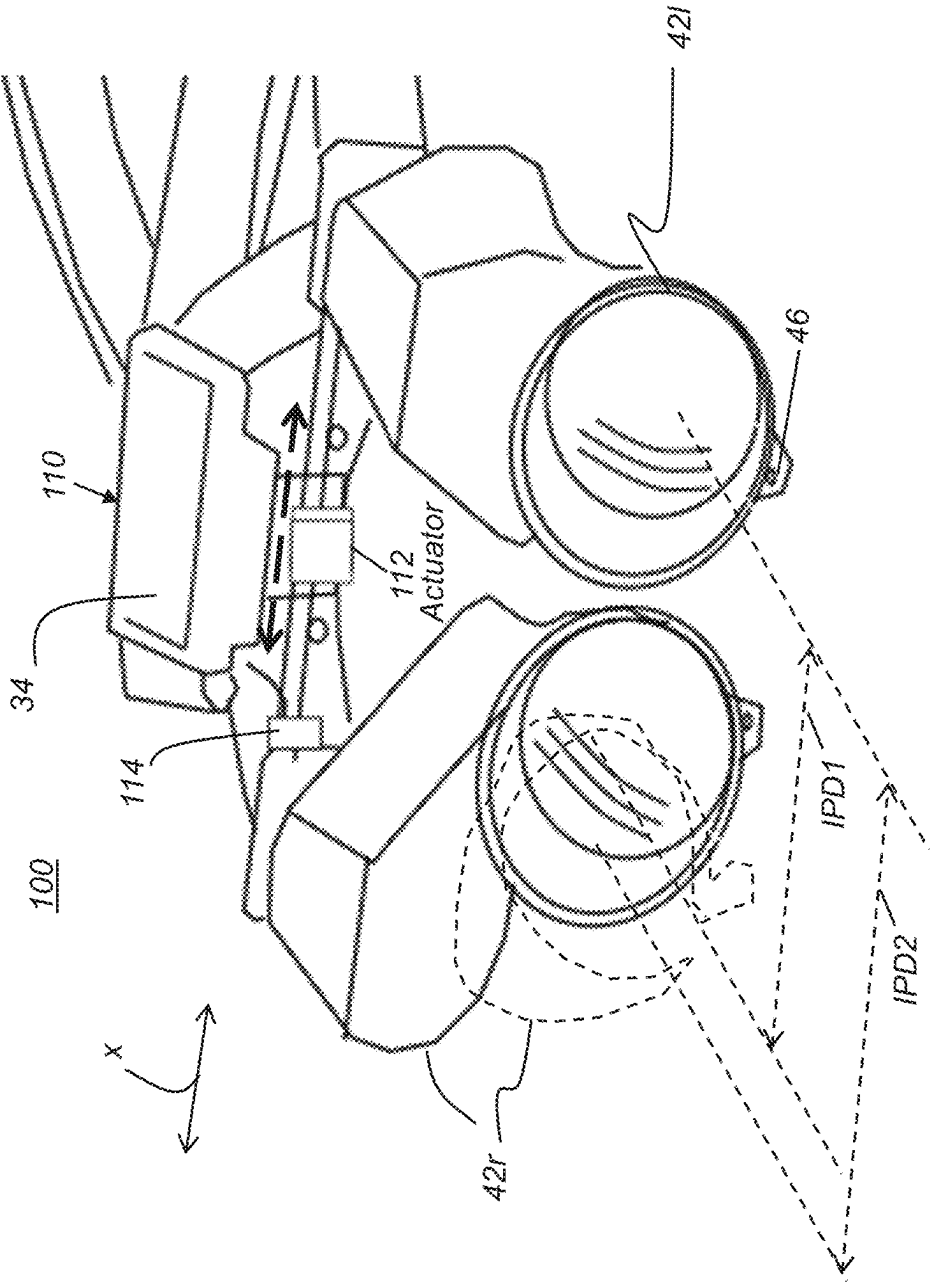


FIG. 10

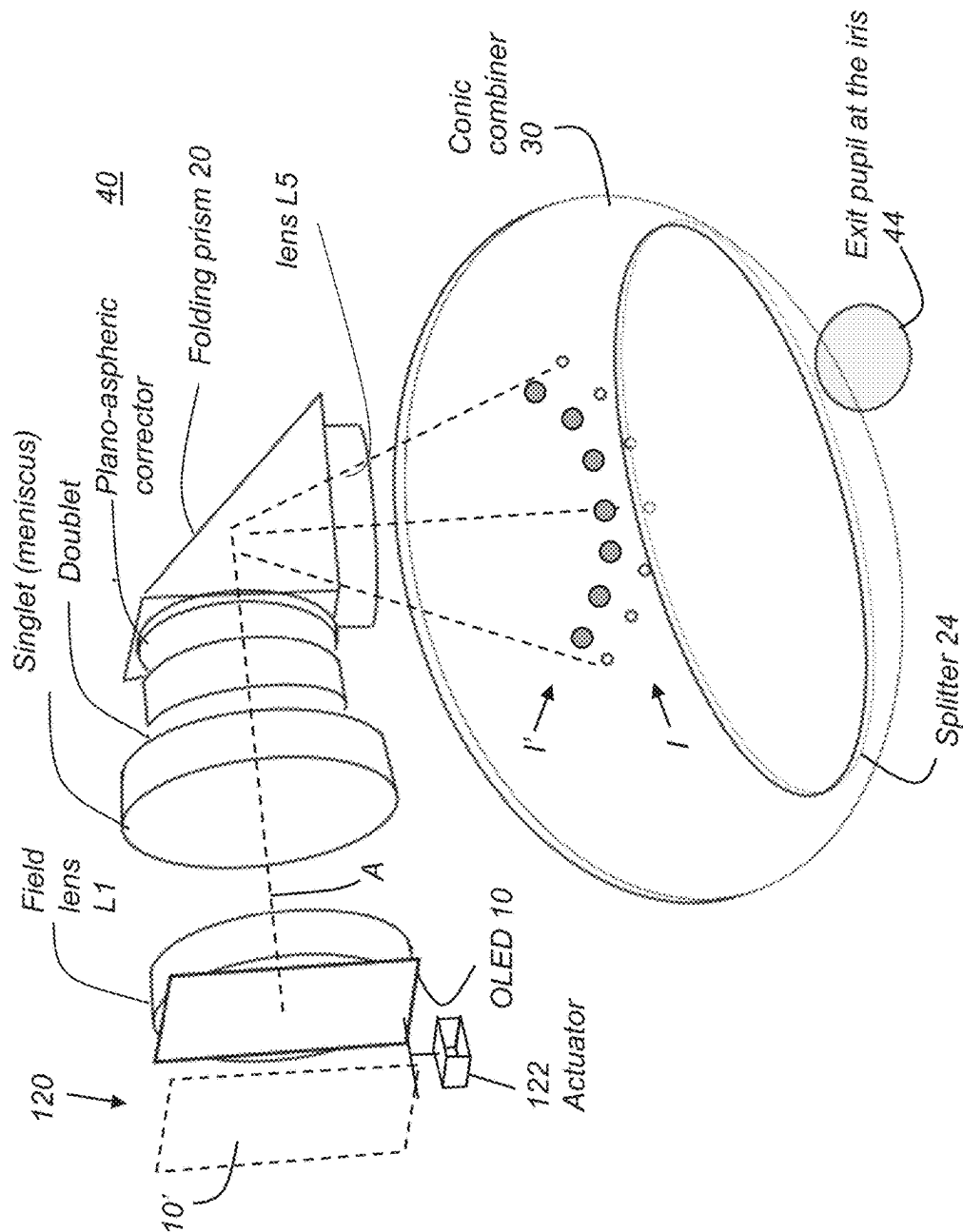


FIG. 11A

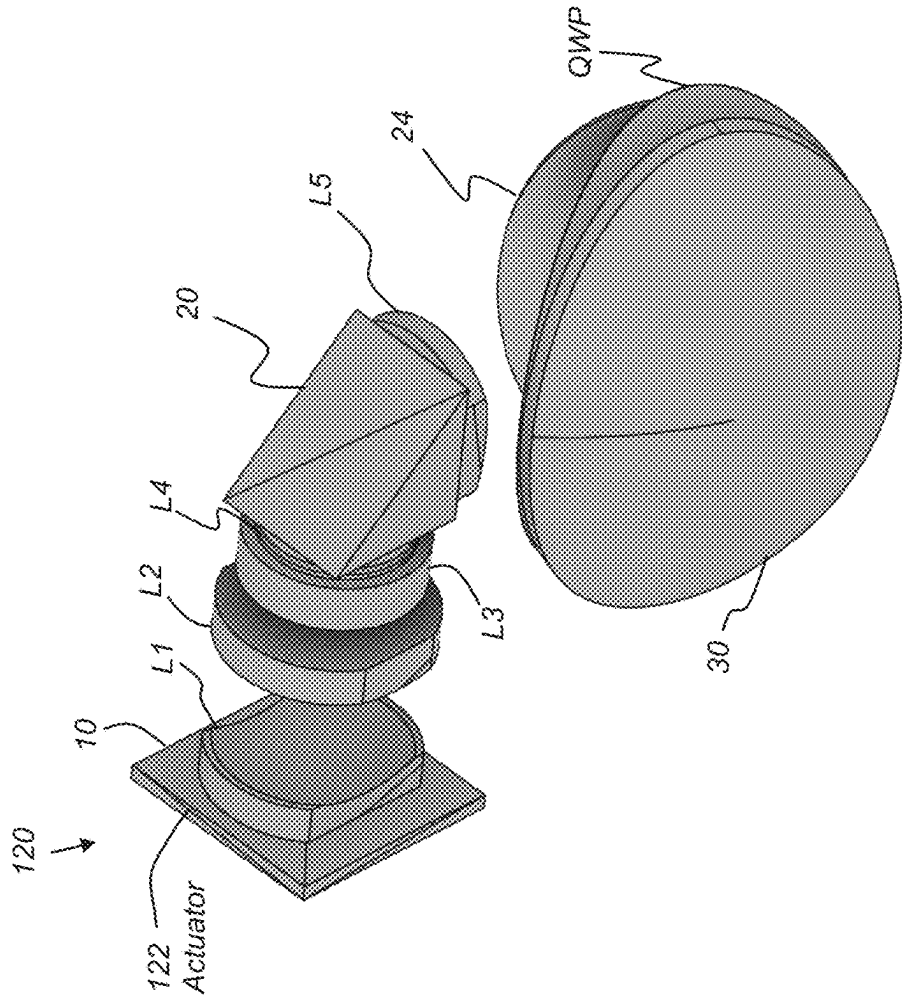


FIG. 11B

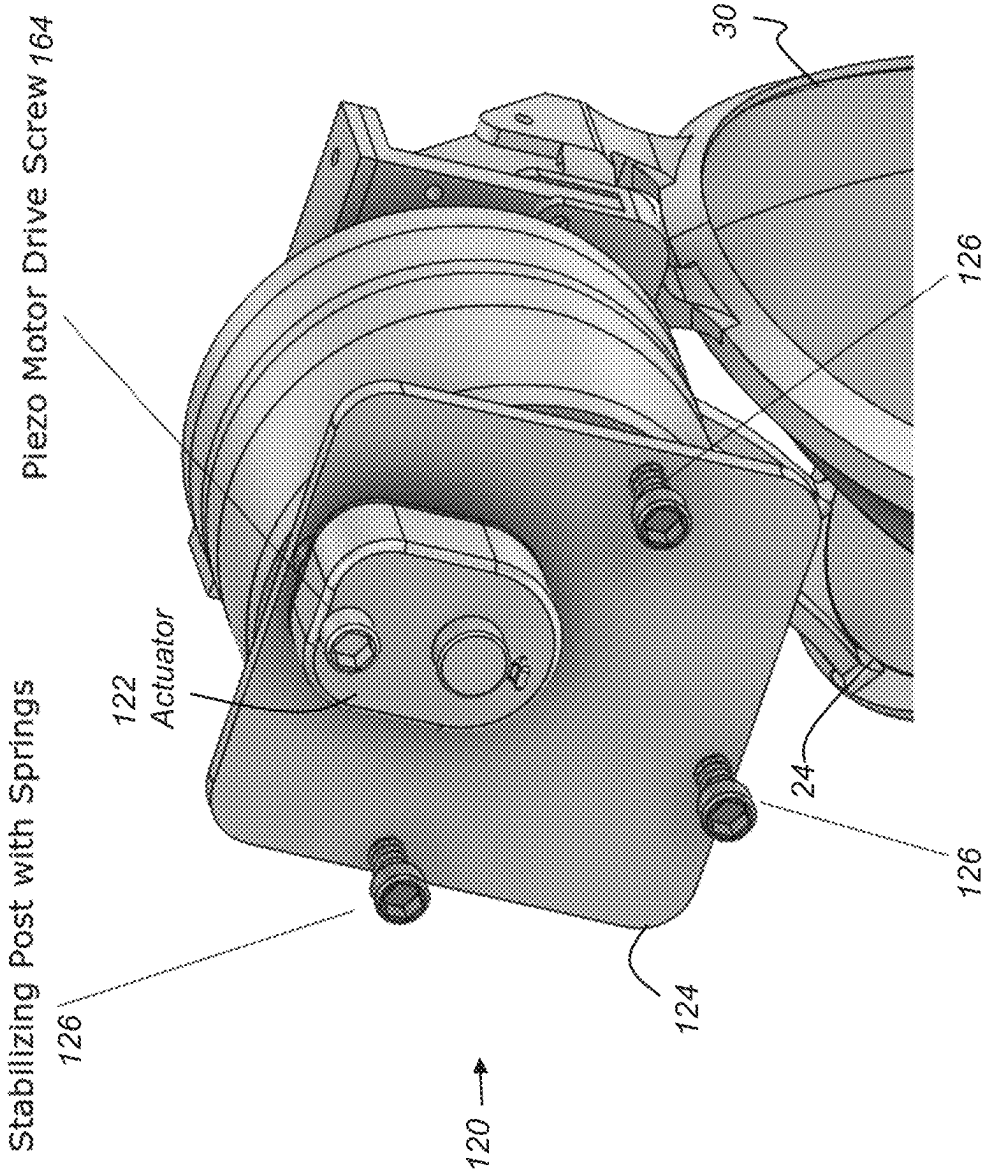


FIG. 11C

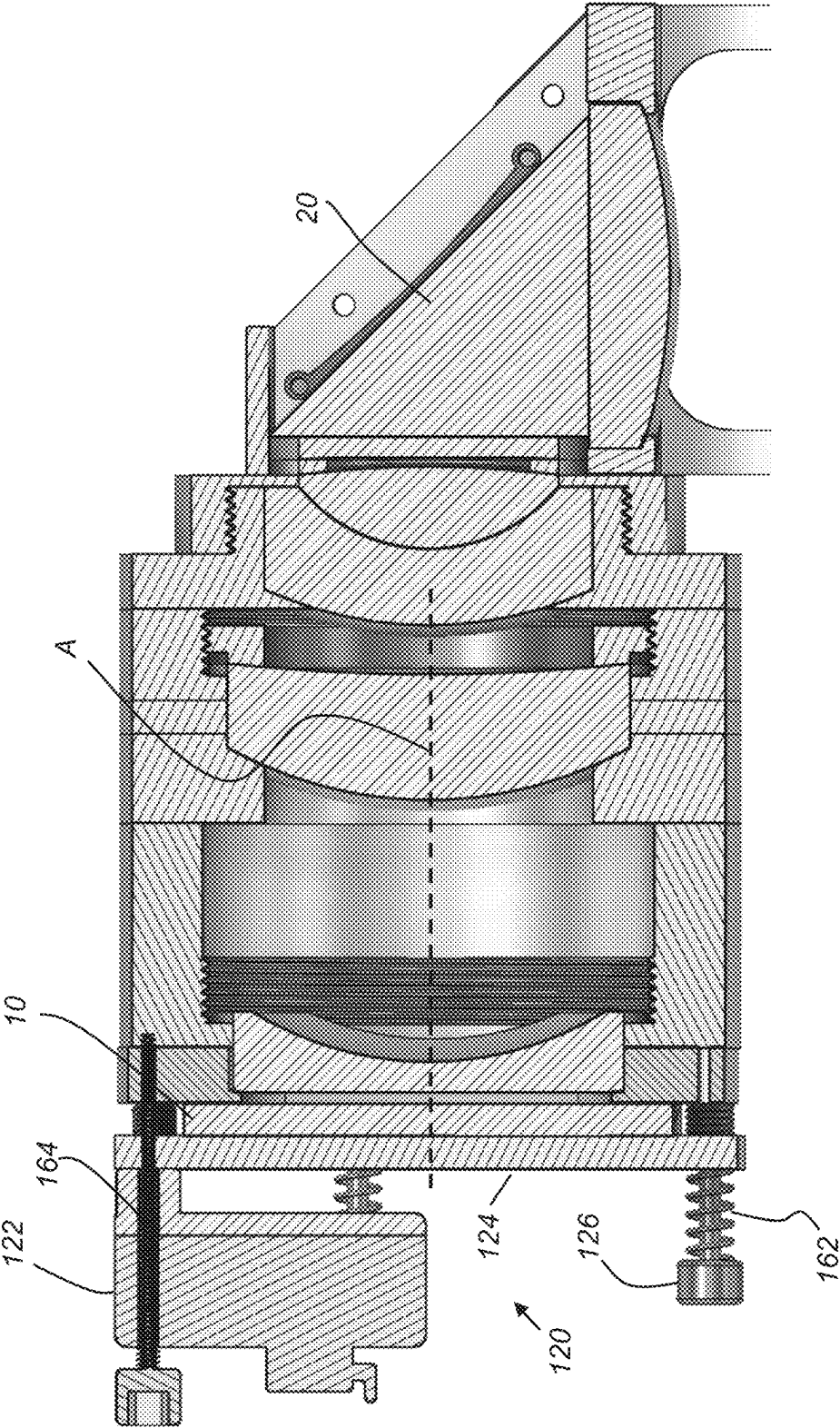


FIG. 11D

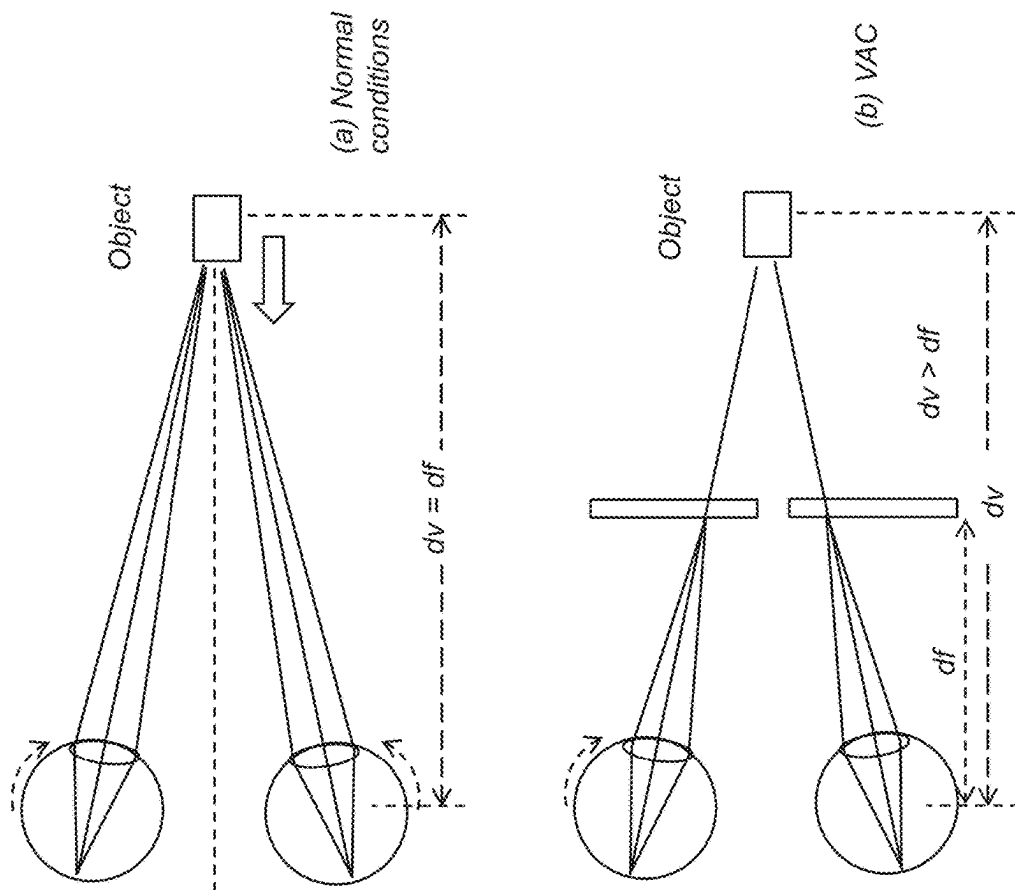


FIG. 12

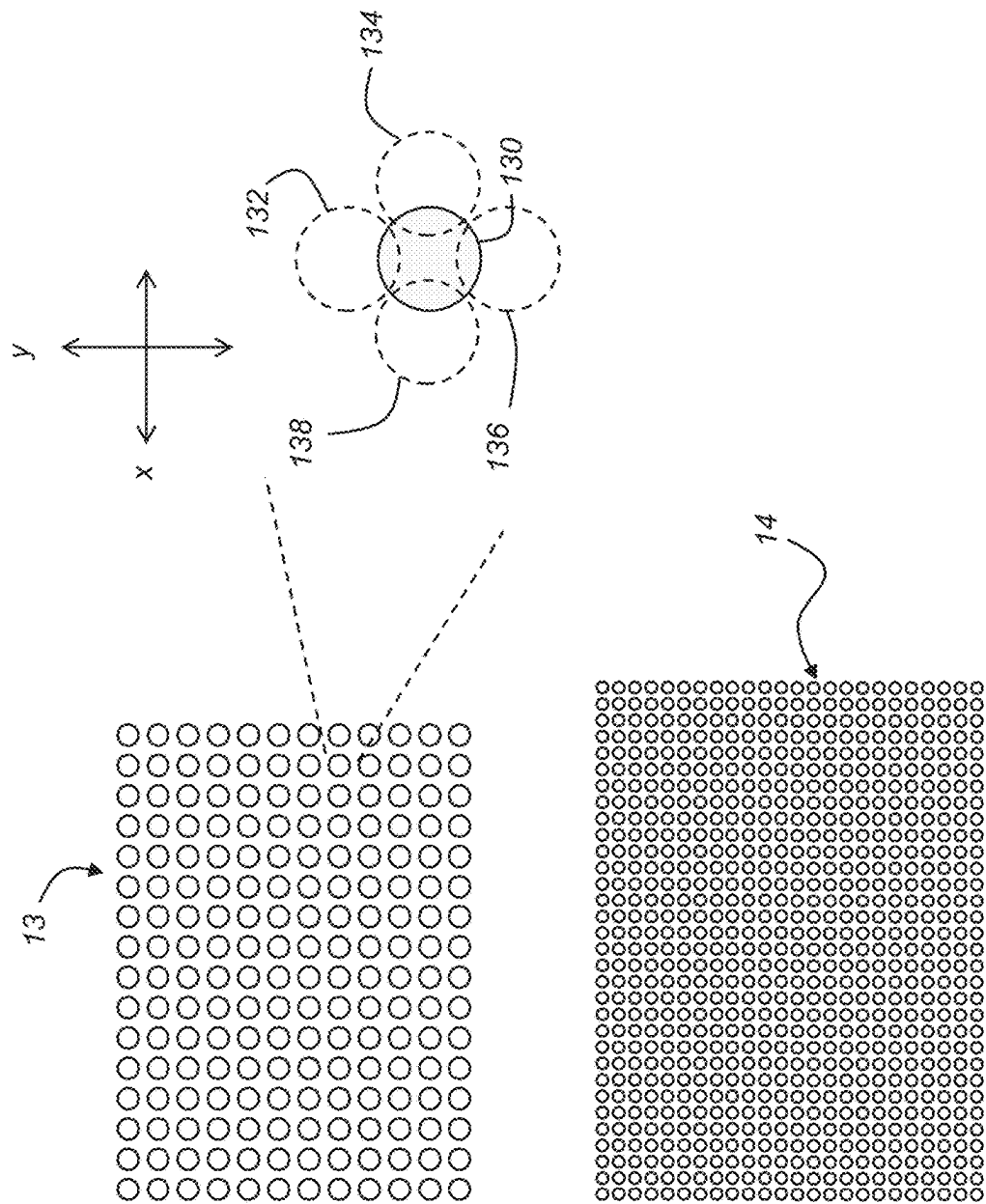


FIG. 13

1

WEARABLE PUPIL-FORMING DISPLAY APPARATUS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/138,240, filed Dec. 30, 2020, which claims the benefit of U.S. Provisional Application Ser. No. 63/060,343, filed Aug. 3, 2020, the disclosure of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

The present disclosure generally relates to wearable display apparatus and more particularly to a wearable display device that provides augmented reality (AR) viewing with a catadioptric pupil-forming optical system that renders a binocular 3D virtual image from a pair of 2-dimensional (2D) displays.

BACKGROUND OF THE INVENTION

Virtual image display has advantages for augmented reality (AR) presentation, including providing the capability for display of image content using a compact optical system that can be mounted on eyeglasses or goggles, generally positioned very close to the eye (Near-Eye Display) and allowing see-through vision, not obstructing the view of the outside world. Among virtual image display solutions for AR viewing are catadioptric optics that employ a partially transmissive curved mirror for directing image-bearing light to the viewer's eye and a partially reflective beam splitter for combining light generated at a 2D display with the real-world visible scene which forms a 3D image when viewed binocularly.

Vision correction applications have employed wearable display devices in order to enhance or compensate for loss of vision over portions of a subject's field of view (FOV). Support for these types of applications can require additional components and can introduce various factors related to wearability and usability that contribute to the overall complexity of the optical design and packaging.

Among challenges that must be addressed with wearable AR devices is obtaining sufficient brightness of the virtual image. Many types of AR systems, particularly those using pupil expansion, have reduced brightness and power efficiency. Measured in NITS or candelas per square meter (Cd/m²), brightness for the augmented imaging channel must be sufficient for visibility under some demanding conditions, such as visible when overlaid against a bright outdoor scene. Other optical shortcomings of typical AR display solutions include distortion, reduced see-through transmission, small eye box, and angular field of view (FOV) constraints.

Some types of AR solution employ pupil expansion as a technique for enlarging the viewer eye-box. However, pupil expansion techniques tend to overfill the viewer pupil which wastes light, providing reduced brightness, compromised resolution, and lower overall image quality.

Challenging physical and dimensional constraints with wearable AR apparatus include limits on component size and positioning and, with many types of optical systems, the practical requirement for folding the optical path in order that the imaging system components be ergonomically disposed, unobtrusive, and aesthetically acceptable in appear-

2

ance. Among aesthetic aspects, compactness is desirable, with larger horizontal than vertical dimensions.

Other practical considerations relate to positioning of the display components themselves. Organic Light-Emitting Diode (OLED) displays have a number of advantages for brightness and overall image quality, but can generate perceptible amounts of heat. For this reason, it is advisable to provide some distance and air space between an OLED display and the skin, particularly since it may be necessary to position these devices near the viewer's temples.

Still other considerations relate to differences between users of the wearable display, such as with respect to inter-pupil distance (IPD) and other variables related to the viewer's vision. Further, problems related to conflict between vergence depth and accommodation have not been adequately understood or addressed in the art.

It has proved challenging to wearable display designers to provide the needed image quality, while at the same time allowing the wearable display device to be comfortable and aesthetically pleasing and to allow maximum see-through visibility. In addition, the design of system optics must allow wearer comfort in social situations, without awkward appearance that might discourage use in public. Providing suitable component housing for wearable eyeglass display devices has proved to be a challenge, making some compromises necessary. As noted previously, in order to meet ergonomic and other practical requirements, some folding of the optical path along one or both vertical and horizontal axes may be desirable.

SUMMARY OF THE INVENTION

The Applicants address the problem of advancing the art of AR display and addressing shortcomings of other proposed solutions, as outlined previously in the background section.

The Applicants' solution uses pupil forming and can be distinguished from pupil expansion systems known to those skilled in the art. By comparison with pupil expansion approaches, the Applicants' approach yields a more efficient optical system with improved image quality. Moreover, the eyes of the viewer can clearly see and be seen by others, with minimal impediment from the optics that provide the electronically generated virtual image.

Wearable display apparatus of the present disclosure are well-adapted for systems that complement viewer capabilities, such as where a viewer may have visual constraints due to macular degeneration or other condition of the eyes.

With these objects in mind, there is provided a wearable display apparatus comprising: (a) an image generator that forms a 2D image, and when worn binocularly, forms a 3D image to the viewer; (b) a partially transmissive mirror having a curved reflective surface; (c) a beam splitter disposed to reflect light toward the curved mirror surface; and (d) an optical image relay that is configured to relay the formed 2D image at the image generator to a curved focal surface of the partially transmissive mirror, wherein the curved focal surface is defined between the curved reflective surface of the partially transmissive mirror and the beam splitter, wherein the optical image relay comprises: (i) a prism having an input surface, an output surface, and a folding surface extending between the input and output surfaces and configured for folding an optical path for light generated by the image generator, wherein an aperture stop for the relay lies within the prism; (ii) a first plano-aspheric lens in optical contact against the prism input surface and configured to guide light from the image generator toward

the folding surface; (iii) a second plano-aspheric lens in optical contact against the prism output surface and configured to direct the light to the curved focal surface; wherein the relay, curved mirror, and beam splitter are configured to form an exit pupil for viewing the 2D image superimposed on a portion of a visible object scene.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1A is a schematic front view showing placement of optical components of the system.

FIG. 1B is a schematic side view showing placement of optical components of the system.

FIG. 2A is a schematic front view showing the optical path through components of the system.

FIG. 2B is a schematic side view showing the optical path through components of the system.

FIG. 3A is a schematic that shows, in perspective view, components of an optical apparatus for AR viewing.

FIG. 3B is a simplified schematic of FIG. 3A.

FIG. 4 is a schematic that shows, from an alternate perspective view, components of an optical apparatus for AR viewing.

FIG. 5 is a side view schematic of an image relay.

FIGS. 6A, 6B are schematic, showing the image relay and components for forming the exit pupil.

FIG. 6C is a schematic view that shows an alternate embodiment of the present disclosure that employs the imaging path for eye tracking.

FIGS. 7A-7C show various features of an embodiment useful for compensating for macular degeneration.

FIG. 8 is a block diagram illustration showing integrated components for an HMD system in an exemplary embodiment.

FIG. 9 is a perspective view that shows positions of various sensors, processor, and components of the head-mounted display according to an embodiment.

FIG. 10 is a schematic diagram that shows components of an interpupil distance adjustment system for the head-mounted display.

FIG. 11A is a schematic diagram showing optical relay components used for focal plane adjustment.

FIG. 11B is a perspective view that shows position of an actuator relative to the display and to corresponding optics.

FIG. 11C is a perspective view from behind the display component, showing a piezoelectric actuator mounted to a plate behind the display.

FIG. 11D is a side view showing components of a dynamic focus adjustment apparatus.

FIG. 12 is a schematic diagram showing visual accommodation and vergence-accommodation conflict.

FIG. 13 is a schematic diagram that shows aspects of dithering geometry for enhanced image resolution.

DETAILED DESCRIPTION OF THE INVENTION

The following is a detailed description of the preferred embodiments of the disclosure, reference being made to the drawings in which the same reference numerals identify the same elements of structure in each of the several figures.

While the devices and methods have been described with a certain degree of particularity, it is to be noted that many modifications may be made in the details of the construction and the arrangement of the devices and components without departing from the spirit and scope of this disclosure. It is understood that the devices and methods are not limited to

the embodiments set forth herein for purposes of exemplification. It will be apparent to one having ordinary skill in the art that the specific detail need not be employed to practice according to the present disclosure. In other instances, well-known materials or methods have not been described in detail in order to avoid obscuring the present disclosure.

Reference throughout this specification to “one embodiment,” “an embodiment,” “one example,” or “an example” means that a particular feature, structure, or characteristic described in connection with the embodiment or example is included in at least one embodiment of the present disclosure. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” “one example,” or “an example” in various places throughout this specification are not necessarily all referring to the same embodiment or example. Furthermore, the particular features, structures, or characteristics may be combined in any suitable combinations and/or sub-combinations in one or more embodiments or examples.

In the context of the present disclosure, the term “eyebbox” has its conventional meaning in the HMD arts, as functionally equivalent to “eye motion box” and similar phrases. The eyebbox is that volume of space within which the viewable image is formed by an optical system or visual display. When the viewer’s pupil is within this volume, the viewer can see all of the generated display content; with the pupil is outside of this volume, the user is typically not able to view at least some portion of the display content. A larger eyebbox is generally desirable, as this allows for lateral and axial movement of the eye, while still maintaining a full field of view. The size of the eyebbox relates directly to the size of the exit pupil for a display system.

Several (or different) elements discussed herein and/or claimed are described as being “coupled,” “in communication with,” “integrated,” or “configured to be in communication with” or a “system” or “subsystem” thereof. This terminology is intended to be non-limiting and, where appropriate, be interpreted to include, without limitation, wired and wireless communication using any one or a plurality of a suitable protocols, as well as communication methods that are constantly maintained, are made on a periodic basis, and/or made or initiated on an as-needed basis.

Where they are used, the terms “first,” “second,” and so on, do not necessarily denote any ordinal, sequential, or priority relation, but are simply used to more clearly distinguish one element or set of elements from another, unless specified otherwise.

In the context of the present disclosure, the term “coupled” is intended to indicate a mechanical association, connection, relation, or linking, between two or more components, such that the disposition of one component affects the spatial disposition of a component to which it is coupled. For mechanical coupling, two components need not be in direct contact, but can be linked through one or more intermediary components.

An embodiment of the present disclosure provides AR viewing and display having a large FOV with an optical system having an optical path that folds in the horizontal or x-direction, the direction substantially parallel (+/-15 degrees) to a line between left and right pupils of a viewer, for forming an intermediate image to the curved mirror. An embodiment of the AR system of the present disclosure has a component arrangement as shown schematically in the front view of FIG. 1A and from the side in FIG. 1B. The corresponding light path is shown schematically in FIGS.

5

2A and 2B, respectively. A flat-panel display is energized as an image generator **10** to form an image and to direct image-bearing light through beam-shaping optics and to a folding prism **20** that redirects the image-bearing light towards a beam splitter **24** and to a curved mirror **30** for forming the virtual image from electronically generated image content. Image generator **10** can be a display that emits light, such as an organic light-emitting device (OLED) array or a liquid crystal array or a micro-LED array with accompanying lenslets, or some other type of spatial light modulator useful for image generation.

In order to address the need for improved overall imaging performance, wider FOV, compactness, and other factors outlined in the background, embodiments of the present disclosure have a number of features shown particularly in FIGS. 3A, 3B, and 4. Specific features of interest include: (i) relay of the image generator **10** to form a curved intermediate image I as a conjugate image. As a type of “aerial” image, intermediate image I is formed in air, serving as the optical “object” for forming the virtual image. Intermediate image I is formed along the curved focal surface of curved mirror **30**, with the approximate aerial position shown by a dashed line in FIG. 5. An optical relay **40**, with particular structure as described in more detail subsequently, conjugates the image formed from image generator **10** to the curved intermediate image I along the focal surface. Curved mirror **30** is partially transmissive, such as between about 30% to 70% transmissive, for example, allowing visibility of the real-world object scene to the viewer. A nominal transmission range of about 50% is useful in many applications. (ii) use of a cylindrically curved quarter-wave plate (QWP) between mirror **30** and beam splitter **24**. Curvature of this element helps to reduce variations of the retardation imparted to the image-bearing light by the QWP over the field of view. (iii) large exit pupil **44**. System optics can form a 10 mm exit pupil at the viewer’s eye-box for eye E. Forming a suitably sized pupil for the viewer helps to provide an eyebox of reasonable dimensions to allow eye movement, without noticeable vignetting. Also, an enlarged eyebox permits the headset to move or slip without noticeable degradation of the viewed image(s). The apparatus does not need to provide pupil expansion, such as is used in existing wearable display apparatus, but uses pupil-forming optics for improved efficiency and brightness, as well as for improved image resolution.

Significantly, the eyes of the viewer can clearly see and be seen by others, with minimal impediment from the beam splitter and curved mirror optics that provide the electronically generated virtual image.

With the optical arrangement shown, the aperture stop AS lies within prism **20** of the image relay, along or very near the fold surface that is provided. This arrangement is advantageous for component packaging and spacing, allowing prism **20** to be reduced in size over other configurations using a folding prism.

The given design allows an FOV along the horizontal (x) axis, the axis parallel to a line between left and right pupils of the viewer’s eyes, of greater than 50 degrees. The FOV aspect ratio (horizontal:vertical) equals or exceeds 1.5. Digital correction is not needed for distortion or lateral color.

According to an embodiment, curved mirror **30** has a conic surface shape. The conic shape is advantaged, in the embodiment shown herein, helping to control chief ray angles, thus correcting for distortion.

Depending on whether or not polarization is used for configuring light paths, beam splitter **24** can be either a polarization-neutral beam splitter or a polarization beam

6

splitter. Beam splitter **24** can be, for example, a wire grid polarization beam splitter as shown in FIG. 4.

Image relay **40**: FIG. 5 shows an enlarged side view of relay **40** for relay of the display image from an electroluminescent display (such as an OLED in the examples shown) to the focal surface position of mirror **30** (shown in a perspective view in FIG. 4) and for shaping the relayed image to suitable curvature to correct distortion. A concave-plano field lens L1, with sides truncated along the vertical direction as shown in FIG. 4 in order to reduce weight and provide a more compact system, shapes the light from OLED display image generator **10**, providing a beam to a meniscus singlet lens L2.

From lens L2, the imaging light goes to a doublet L3 having a concave/convex flint glass lens cemented to a crown glass lens.

An aspheric plano-convex lens L4 is in optical contact with the input face of prism **20**, such as cemented to prism **20**. A second plano-aspheric lens L5 can be cemented to the output face of prism **20**. This cemented arrangement facilitates alignment of these optical components. According to an alternate embodiment, only a single plano-aspheric lens L4 is deployed at the prism **20** input surface.

The turning surface **26** of prism **20** is coated to enhance reflection. Hypotenuse or turning surface **26** of the prism is essentially the relay (and system) aperture stop.

Intermediate image I is formed in the shape and location of the focal surface of the curved mirror. Proceeding backward along the optical path from intermediate image I are the following components: Plano-asphere lens L5; Folding prism **20** with turning surface **26**; plano-asphere lens L4; Doublet L3; Meniscus singlet L2; Field lens L1; Image source or generator, display **10**.

FIG. 6A and simplified FIG. 6B show an alternate view of the display optics from exit pupil **44**. Chief rays are shown in FIG. 6B; these chief rays converge at the position of exit pupil **44** at eye E. FIG. 6B also shows the approximate position of an intermediate pupil P_i at the aperture stop, near the folding surface **26** of prism **20**.

As shown in FIGS. 1A and 4, the image generator is disposed to direct image-bearing light beam in a horizontal direction and along a path that lies above eye-level, as the display optics are normally worn by a sitting or standing viewer. Prism **20** can be tilted slightly away from the forehead of the viewer, to direct light in front of the face at an oblique angle to vertical, as shown in the embodiment of FIG. 2B.

The layout and routing of the optical path are particularly suitable for providing augmented reality 2D and 3D viewing in a wearable device. Using relay **40** allows the positioning of image generator **10** to be out of the direct field of view; in addition, relay **40** allows image generator **10** to be positioned at sufficient distance away from the skin surface to avoid contact and consequent discomfort. The use of a first x-direction (horizontal) fold, followed by a y-direction (vertical) folding enables the imaging optics to be compactly packaged with reasonable optical path distance to allow a measure of light beam shaping and correction. Prism **20** can be rotated over at least a range of angles about the x axis, allowing a measure of alignment as well as adaptation to different mirror **30** curvatures. Employing a curved surface for an optional QWP component helps to reduce variations, over the FOV, of retardation imparted by the QWP; excessive variation over the field may otherwise cause some brightness fall-off.

Using a wire-grid polarizer reduces light loss, allowing high levels of visibility to the external, real-world object scene content, along with reduced light leakage over other polarization components.

Image source **10** may be unpolarized. In one embodiment, a polarizing beam splitter is used, such as a wire grid splitter made by Moxtek, Inc., Orem, UT. This type of beam splitter reflects only one polarization, usually S polarization, towards the conic combiner. The orthogonal polarization, P polarization, is transmitted and is absorbed (absorber not shown). To prevent the small amount of P light from being reflected, an optional polarizer can be placed at the image source.

The mirror **30** provides a conic combiner in embodiments shown, with power only for the generated image and not for the visible field. The curved mirror **30** can be a double conic for improved image formation. Various types of coatings can be provided on the mirror **30** combiner, including, but not limited to dichroic coatings, metal coatings, such as to provide a half-silvered reflector, electrochromatic coatings, anti-reflection (AR) coatings. Mirror **30** can be fully or partially reflective or fully or partially transparent, with some amount of reflectivity.

Embodiments of the present disclosure provide a measure of distance between the image generator (OLED or other spatial light modulator device) and the face and temples of the viewer. This helps to prevent discomfort due to heat where the wearable display is worn for an extended period of time.

The particular arrangement of image-forming components provides suitable image quality and high resolution to allow reading and other visual activity involving fine detail.

According to an embodiment of the present disclosure, the optical system described herein is suitable for applications requiring sensitivity to the viewer, including not only viewer comfort, but some level of vision monitoring and adaptation. For example, the apparatus described herein can be used as part of a system for compensating for vision problems. By way of example, FIGS. **6C** and **7A-7C** show various features of an embodiment useful for compensating for macular degeneration. This type of application can require a measure of viewer monitoring and adaptation, possibly including adjustment of generated data content suitable for the viewer.

Gaze tracking: According to an aspect of the present disclosure, gaze tracking can be provided as part of the wearable optics system and used to adjust system parameters according to perceived focus of attention for the viewer. Cameras and infrared (IR) light sources provided on a headset, as shown subsequently, can provide the gaze-tracking function. Gaze tracking can be combined with the controller and with a camera image FOV intake. For instance, change of the image aspect ratio may be appropriate, allowing the system to adapt image content to the dimensional parameters available from the image generation system. Thus, for example, cameras associated with the HMD can oversample the real-world input from the object scene, acquiring a wider FOV than can be displayed on the lens. The resulting buffered images are related to the reduced FOV video that can be projected, as controlled by using the sector of the FOV identified using eye gaze recognition.

FIG. **6C** is a schematic view that shows an alternate embodiment of the present disclosure that employs the imaging path itself for eye tracking and provides 1:1 imaging of the viewer's iris. IR light, or other sensing light, is directed along the optical path, such as generated from, through, or at some other point along the output path of,

image generator display **10**. Folding surface **26** can be formed as a dichroic surface, treated to direct the sensing light to beam splitter **24** and to curved mirror **30** and to the iris of eye **E**. Sensor light returning from the iris generally retraces the light path to prism **20**. A portion of this returned light from the viewer's eye, as shown by a dashed line in FIG. **6C**, can be transmitted, rather than reflected, through surface **26** and be conveyed through a complementary facing prism **60** to a tracking sensor **62**. This returned light can be IR light, for example. The IR light source can be directed to the eye from any suitable position. Surface **26** is then configured to reflect most of the visible light, performing its turning function for image-bearing light, and to transmit IR light suitably for gaze tracking.

Other configurations are possible. Thus, for example, a dichroic coating can be employed for surface **26**, or some other coating can be employed that provides the needed redirection by reflection of the bulk of image-bearing light, while also allowing sufficient light leakage for sensing.

The embodiment of FIG. **6C** can be modified in a number of ways to allow eye gaze tracking through portions of the imaging path. For example, with corresponding changes to components, the use of IR light as a sensed light can be replaced by employing visible light. This would require surface **26** to be partially reflective in the visible range, such as 90% reflective and 10% transmissive, for example, wherein the transmitted light is the sensed portion. One or more additional lenses, not shown in FIG. **6C**, could be provided in the path of light to sensor **62** for gaze tracking using the image path as described herein.

Headset configurations: FIGS. **7A-7C** show a stereoscopic display headset **100** that has separate optical paths for the left and right eyes of a viewer. As particularly shown in FIGS. **7A** and **7C**, the optical relay **40** can be compactly packaged as part of a headset **100** within an optical module **42** that is disposed above eye level, as the system is normally worn. Beam splitter **24** and curved mirror **30** are arranged to lie along the visual axis of the viewer. An adjustable strap **102** can be provided to allow adaptation to viewer anatomy. A control processor **34** can include the needed electronics for controlling operation of the optical apparatus; processor **34** components can be mounted above the visual axis and disposed away from the viewer's forehead.

Eye-tracking can be provided from the headset using the arrangement previously described with respect to FIG. **6C** or using one or more eye-tracking cameras **46**, working in conjunction with illumination LEDs **48**, typically infra-red (IR) sources, as shown in FIG. **7B**.

FIG. **8** shows a headset **100** that incorporates the optical system described herein as part of a wearable system for providing user information and guidance in performing a task or assignment. Optical relay **40** and its associated components are packaged within an optical module **42**. A variety of sensors **80**, described in detail subsequently, can include one or more SLAM sensors **82**, as well as sensors for various environmental or ambient conditions such as temperature, humidity, and the like, viewer-related, contextual data, cameras, and other information-gathering devices can be integrated with or in wireless or wired signal communication with control logic and related processing components of the HMD headset. Sensors can be provided to support AR imaging, VR imaging, or mixed AR/VR imaging modes of operation.

Connection to power and to signal sources for the headset can be obtained by connection of the headband of headset

100 with external power and signal sources, such as other computing and processing equipment worn or carried by the viewer.

System block diagram: FIG. 8 is a diagrammatic block diagram illustration showing, for an exemplary embodiment, interrelation of integrated components for an HMD system including input sensors 80, output components and systems 90, logic and control components including processors, graphic processing units (GPU), MVC and other devices. Sensors 80 can include high-resolution cameras, multiple displays per eye, 6 to 9 degrees of freedom sensor or other sensors such are necessary for detection of hand-gesturing, head-gesturing, voice control, positional location, and estimation or navigation, as well as optical character recognition (OCR), tracking, marker-based or markerless-based AR, location, SLAM sensors, concurrent odometry and mapping sensors, microphones and noise-cancelling microphones, and any other sensors which could be used on an AR/VR headset. Previous figures, for example, gave a diagrammatic illustration of a placement of an IR light for the eye-tracking subsystem. The perspective view of FIG. 9 shows positions of various sensors, processor, and components.

Among other sensor technologies which may be housed on the HMD are manual control inputs. These can include digital buttons, which may include power buttons, and a D-Pad or control-pad for accessing and controlling functions by the user, which may or may not be in a dongle; and if not in a dongle then it may exist on the headset or in a wired or wireless remote control. The sensors listed above may include their operating systems and output. The control mechanism may also respond to other types of input, including voice command, SLAM, eye tracking, head or hand gesturing, or any other method which can be employed with the sensors and systems mentioned above.

HMD 100 may also house connectors such as power connection for recharging a battery or for direct connection to an AC source, for the HMD as well as for related input and output devices. There can also be additional external connectors for HDMI, sound, and other input/outputs, such as additional image overlay display, or for a diagnostics protocol for upgrading the system. The HMD may also house the microprocessor(s) control circuits. HMD 100 may also contain one or more display per eye, allowing the use of any number of additional projectors, like Pico projectors, or micro-displays. The displays may be used to project through either catoptric system, a dioptric system, or catadioptric system, or combinations thereof, such as to generate an ultra-short-throw image onto reflective lenses or to project to some other surface, which can be clear plastic, like a polycarbonate resin thermoplastic (Lexan).

The HMD 100 may also house a rechargeable battery which is not typically removed, thus, providing spare energy to continue to power the HMD when the removable battery is exhausted or removed. While this battery may be smaller and only have a run-time of several minutes, it can provide the HMD with a "hot-swap" battery system that permits a user to keep viewing from the HMD for a time after the removable battery has died or been removed.

HMD 100 may also include a strap and counterweight or other headgear to balance the HMD and maintain its position on the head. The HMD may contain a "pinch adjustor" to adjust strap 102. In addition, the HMD may or may not include a "dongle" whereby one or more of the systems or subsystems may be connected via wired or wireless to another device, such as could be worn on a belt or carried in a pocket to reduce the overall weight of the HMD 100. In

one embodiment, the HMD may be connected to another device which is providing power, while in an alternative embodiment, the HMD may have its own power from the mains or from wireless power transmission or from a battery.

Further, in another embodiment, the HMD may house other subsystems such as the cameras, the microcontrollers, the connectors, central processing unit, graphics processing unit, software, firmware, microphones, speakers, display, and collector lens; the displays, and other subsystems.

In another embodiment, the HMD may contain a front facing sensor array along with other sensors mentioned above and optical character recognition (OCR) and/or cameras to read and/or measure information from the real world object scene. Additionally, the HMD may contain one or more connectors to connect via wire to the outside world for power and data (i.e. USB, HDMI, MiniUSB).

Alternatively, some parts of the system mentioned herein may be in a dongle attached to the HMD via wire or wireless connection. Alternatively, some portions of the system mentioned herein may be contained in a connected device, like a laptop, smart phone, or Wi-Fi router. Alternatively, some parts of the system mentioned herein may be contained in a smartphone or may be transferred back and forth from a smartphone to the HMD, when synced, such as the HMD displaying the smartphone apps and other features of the smartphone that would otherwise be displayed on the smartphone display. Alternatively, the HMD may contain and display all the features that a smartphone can.

In another aspect of the present disclosure, HMD 100 may contain all the features of a typical smartphone and no connection may be needed with a smartphone to have all the smartphone features, like web or cell calling, app use, SMS, MMS, or similar texting, emailing, logging on to the internet, and the like.

According to an aspect of the present disclosure, the HMD headset may provide a computer mediated video shown on the reflective lens layer such that the wearer may see both the real world and the augmented video at the same time. In this aspect of the disclosure, such features as voice/speech recognition, gesture recognition, obstacle avoidance, an accelerometer, a magnetometer, gyroscope, GPS, special mapping (as used in simultaneous localization and mapping (SLAM)), cellular radio frequencies, Wi-Fi frequencies, Bluetooth and Bluetooth Light connections, infrared cameras, and other light, sound, movement, and temperature sensors may be employed, as well as infrared lighting, and eye-tracking.

Batteries and other power connections may be needed for various devices, but are omitted from schematic figures for clarity of other features.

SLAM sensors: Embodiments of the disclosure can further include mechanisms and logic that provide SLAM (simultaneous localization and mapping) capabilities to support the viewer. SLAM uses statistical techniques to map the viewer's environment and to maintain information on the viewer's relative position within that environment. For example, an image from a Simultaneous Localization and Mapping (SLAM) camera configured for the wearable unit can detect a location of the HMD wearer within the given environment. SLAM capabilities can also be useful where some portion of the viewer's FOV is blocked or otherwise obscured. Using SLAM allows the system to present portions of the real-world object scene in the electronically generated image.

To support SLAM capabilities, a headset as shown in the example of FIG. 9 can include one or more SLAM sensors 82, such as cameras. In some cases, the SLAM camera may

11

include a visual spectrum camera, an infrared (IR) camera, or a near-IR (NIR) camera. Additionally, or alternatively, the SLAM camera may be include or exclude the viewer from its field of view. Techniques for providing SLAM capabilities are known to those skilled in the mapping arts and can include processing of local image content as well as use of tracking data.

SLAM logic can be provided by control processor 34 or by an external processor that is in signal communication with processor 34, including processors that are connected to the wearable display device by a wired connection or, alternately, processors that are in wireless communication with control processor 34.

IPD Adjustment System: Embodiments of the present disclosure address the need for HMD system adjustment to viewer anatomy, with benefits for system efficiency and usability. One aspect of variable viewer anatomy relates to inter-pupil distance (IPD). This well-known characteristic relates to overall head dimensions and position of the eye sockets. Mismatch of IPD by the device can make it difficult to provide image content and needed functions and can make the HMD difficult to wear and use for some viewers. Some systems provide manual methods for IPD adjustment. As shown in the perspective view of FIG. 10, the Applicant's HMD can provide an automated IPD adjustment system 110 that measures the IPD for the viewer and responds by adjusting the inter-pupil spacing, changing the distance between the left and right optical modules 421 and 42r, respectively.

Referring to the FIG. 10 depiction, IPD adjustment system 110 can have an actuator 112 that urges one or both left and right optical modules 421, 42r either farther apart, as shown in distance IPD2, or closer together as in distance IPD1. This adjustment can be performed at power-up or other system activation, at operator command, or at predetermined periods, or at some other time or in response to other event. According to an embodiment of the present disclosure, IPD adjustment settings can be stored for each viewer who has previously used the HMD. Then, upon entry of information identifying the viewer, the stored IPD adjustment settings can be restored as presets.

There may even be an external device, motorized or mechanical, which effects the activation of the preset IPD upon instruction prior to wearing the HMD.

In one embodiment, to execute the IPD adjustment function, eye-tracking cameras 46 on both left and right optical modules 421, 42r obtain image content that allows pupil center detection by processor 34 logic. This logic determines the relative location of actual pupil centers and IP distance and determines whether or not the IPD between pupil centers is compatible with the positioning of left and right optical modules 421, 42r. If positioning is appropriate, no IPD adjustment is necessary. Otherwise, an actuator 112 can be energized to translate one or both left and right optical modules 421, 42r in the horizontal or x-direction as shown in FIG. 10. At various incremental positions, feedback logic can again measure and calculate any needed adjustment until a suitable IPD is achieved. For IPD adjustment, at least one of the left and right optical modules 421 and 42r can be movable; the other module 421 or 42r can also be movable or may be stationary.

Adjustable Focus: As a useful default for most virtual image viewability, HMD optics are typically designed to form the virtual image so that it appears to be at optical infinity, that is, with rays substantially in parallel to the optical axis. In an embodiment, the HMD optics are designed to provide the augmented virtual image at about 1

12

m distance from the viewer eyebox, equivalent to approximately 1 diopter prescription glasses.

In some applications, closer focal length is advantageous. To achieve this, the Applicant's solution provides measurement and adjustment for diopter adjustment of the optical relay 40 optics. Referring to FIG. 11A, there is shown a schematic for relay 40 components that includes an actuator 122 and associated components as part of a focal plane adjustment system 120. By changing image generator 10 position along an axis A, a change in focal position is effected, such as with image generator 10 shifted to the dashed outline position denoted for image generator 10' in FIG. 11A. This movement causes a corresponding shift of intermediate image I to the position shown as image I' in relay 40. The splitter 24 and combiner, curved mirror 30, then condition the image-bearing light to provide a virtual image at a shifted spatial location. Actuator 122 can be a linear piezoelectric actuator, for example, capable of high-speed change between positions. One or more actuators 122 can be used for moving any of the components described hereinabove with relation to optical relay 40 in order to adjust the position of the focal plane.

FIGS. 11B, 11C, and 11D show different views of relay optics with focal plane adjustment system 120. FIG. 11B is a perspective view that shows position of actuator 122 relative to image generator 10 and corresponding optics. FIG. 11C is a perspective view from behind image generator 10, showing a piezoelectric actuator 122 mounted to a stationary plate 124 behind image generator 10 (not visible in the view of FIG. 11C). Actuator 122, such as a piezoelectric actuator, translates image generator 10 along axis A, which extends orthogonally to the electroluminescent display component surface of image generator 10. FIG. 11D shows a side view of adjustment system 120 components. One or more stabilizing posts 126 has a screw and a compression spring 162 for maintaining image generator 10 in position along the optical path, so that movement of image generator 10 is constrained to the axial direction (axis A) during actuation.

One difficulty with the change in focal length relates to vergence-accommodation conflict, as shown in the schematic diagram of FIG. 12. Vergence-accommodation conflict (VAC) is a vision phenomenon that is familiar to developers of three-dimensional (3D) displays and virtual-reality (VR) display devices. Under real-world normal viewing conditions, where only the object scene is in view as shown in part (a) of FIG. 12, the viewer's eyes converge, rotating toward one another to focus on closer objects and, correspondingly, away from one another to focus on objects at further distance. Accommodation, as the process where the lenses of the eyes focus on a close or far away object, is consistent with convergence for normal viewing in the part (a) depiction and involves epipolar geometry of the HMD cameras and the wearer's eyes. Thus, accommodation and convergence can be considered to be coupled.

Focal plane adjustment system 120 (FIGS. 11A-D) also employs eye-tracking sensors and related data for determining when there is a discrepancy between focal planes for the object world FOV and the generated image content. The focal plane position for the generated image can be computed according to system optical geometry for the components defined hereinabove. Later adjustments to the geometry, executed by system logic for shifting focal plane position, can be recorded and used to recalculate focal plane position.

In FIG. 12 part (a), the viewer sees a real object, i.e., the viewer's eyes are verged on the real object and gaze lines

from the viewer's eyes intersect at the real object. As the real object moves nearer the user, as indicated by the arrow, each eye rotates inward (i.e., converges) to stay verged on the real object. As the real object gets closer, the eye "accommodates" to the closer distance, through reflexive, automatic muscle movements, by adjusting the eye's lens to reduce the focal length. In this way, accommodation adjustment is achieved. Thus, under normal viewing conditions in the real world, the vergence distance (dv) equals the accommodation distance (df).

In 3D displays and VR systems, however, these two processes can be decoupled, as shown in part (b) of FIG. 12. FIG. 12 shows an example conflict between vergence and accommodation that can occur with stereoscopic three-dimensional displays, in accordance with one or more embodiments. In this example, an observer is looking at the virtual object displayed on a 3D electronic display; however, the observer's eyes are verged on and gaze lines mapped from the observer's eyes intersect at the virtual object, which is at a greater distance from the observer's eyes than the image formed by the 3D electronic display. As the virtual object from the 3D electronic display is rendered to appear closer to the viewer, each eye again rotates inward to remain verged on the virtual object, but the focus distance of the image is not reduced; hence, the observer's eyes do not accommodate, as in part (a). Thus, instead of increasing the optical power to accommodate for the closer vergence depth, the eye maintains accommodation at a display distance associated with the 3D electronic display. Thus, the vergence depth (dv) often does not equal the focal length (df) for the human eye for objects displayed from 3D electronic displays. This discrepancy between vergence depth and focal length is referred to as "vergence-accommodation conflict" or VAC. A user experiencing only vergence or only accommodation, and not both vergence and accommodation, can eventually experience some degree of fatigue, dizziness, discomfort, disorientation, and even nausea in some cases.

In order to compensate and correct VAC, the FIGS. 11A-D relay optics can adjust the relative position of intermediate image I to I', moving the virtual image focal plane more closely toward the real-world focal plane.

Model controller: Embodiments in accordance with the present disclosure may be provided as an apparatus, method, computer program, hardware/software, state machine, firmware, and/or product. All of the systems and subsystems may exist, or portions of the systems and subsystems may exist to form the apparatus described in the present disclosure. Accordingly, one or more portions of the Applicant's solution may take the form of an entirely or partial hardware embodiment, a predominantly software embodiment (including firmware, resident software, micro-code, etc.), or an embodiment combining software and hardware aspects in some combination that may all generally be referred to herein, without limitation, as a "unit," "module," or "system." Furthermore, one or more portions or functions for the present disclosure may take the form of a computer program product or products embodied in any tangible media of expression or storage having computer-usable program code embodied in or otherwise represented using the media. Any combination of one or more computer-usable or computer-readable media (or medium) may be utilized, including networked combinations that utilize remote processing components. For example, a random-access memory (RAM) device, a read-only memory (ROM) device, an erasable programmable read-only memory (EPROM or Flash memory) device, a portable compact disc read-only memory (CDROM), an optical storage device, and a magnetic stor-

age device. Computer program code for carrying out operations of the present disclosure may be written in any combination of one or more programming languages. Further, the intelligence in the main circuitry may be software, firmware, or hardware, and can be microcontroller based or included in a state machine. The disclosure may be a combination of the above intelligence and memory and this can exist in a central processing unit or a multiple of chips including a central graphics chip. The computer portion of the disclosure may also include a model view controller (MVC) as shown in FIG. 8, which is also called herein a "model controller."

Dithering: According to an embodiment of the present disclosure, dithering can be employed to modify and improve the visual experience of the viewer. Dithering can be effected, for example, by rapid in-plane vibration of a camera or image generator 10 using a piezoelectric actuator 122, as was described previously with respect to FIGS. 11A-D. Dithering, imparted to the displayed image content using synchronous, timed spatial displacement, can be a desirable solution for helping to mask or eliminate display-related artifacts, for example.

Dithering can also be used to enhance image resolution using Timed Spatial Displacement. Improved image resolution is goal, and holds promise for future use of AR/VR glasses in various applications, such as in critical use cases such as surgery visualization. In these applications, for example, detail visualization of fine layers and blood vessels can be critical to successful outcomes. Micro displays continue to mature, with pixel counts of 2 million in a single small device, with further improvements likely. These higher resolution displays impose steeply increased demands on system resources, including higher power and computation speed and complexity, for example.

The Applicant addresses this problem using dithering of display components. This solution can make higher resolution images available to users at a discounted power cost, which in turn can provide lighter, cooler running systems.

Increased pixel resolution is obtained by using the capability to shift image generator 10 in-plane, that is, in one or more directions parallel to the emissive display surface of image generator 10, synchronously with corresponding changes in image data content. With respect to FIG. 11D, image generator 10 translation is orthogonal to axis A for dithering to increase pixel resolution using synchronous, timed spatial displacement. Actuator 122 is configured to provide dithering using in-plane translation of image generator 10.

As shown schematically in FIG. 13, multiplying the image generator 10 resolution is accomplished by physically shifting an array 13 of pixels that form image generator display 10 in the x-y plane. The shift distance is a fraction of the pixel pitch. At right in FIG. 13 is represented a pixel 130 of array 13, shifted in the x-direction to pixel positions 134 and 138 and shifted in the y-direction to positions 132 and 136. Synchronous with the shift action is modulation of image data for each pixel 130 at its original position and at each shifted position 132, 134, 136, and 138. Thus, for example, the state of a pixel at its position 134 can differ from its state at position 138, according to the image data content that is provided with the shift. With a half-pixel shift in each x- and y-direction, for example, the effective pixel count can be increased by at least 4 times. With a half-pixel shift only along one axis, such as only along the x or y axis as shown, the effective resolution along the axis parallel to the shift can be double. Overall, the power cost of nano scale piezo shifting is much lower than the cost to design and

15

implement 4× the pixel count using a higher resolution image generator 10 element. An array 14 represents increased pixel resolution.

For the embodiment of FIG. 13, an image generator 10 was provided, having a 240 frames per second (fps) refresh rate. In terms of the piezoelectric actuation provided, each pixel Element can be Relocated at ½ the Delta of the Pixel Element Center-to-Center Distance in the Array 13. This arrangement can provide 60 fps display at 4X the resolution of the original image generator 10.

By way of example, an embodiment of the present disclosure employs QNP-XY Series Two-Axis, CY Piezo Nanopositioners for image generator 10 dithering actuation.

A number of features allow piezoelectric dithering to provide enhanced resolution, including the following: (1) Adjustable travel range, such as from 100 μm to 600 μm, for example; (2) Long device lifetimes; (3) Superior positioning resolution; and (4) High stiffness and other factors.

A number of piezoelectric actuators provide the option of closed-loop feedback that allows sub-nanometer resolution and high linearity.

Dithering for increased resolution can utilize any of a number of movement patterns for in-plane displacement of pixels. Patterns for dithering the pixels in the display (or the displays) for increased resolution include transposing with rectilinear motion, curvilinear motion, or in a translational, rotational, periodic, or non-periodic motion, or any of the combination of the above. Also, the pattern could include a rectangle pattern which may increase the resolution by 4 times. Another alternative way to address the pixel movement is to have pixels that are approximately the same size as the non-emissive dark or “black” space between the pixels where the dithering translation of each pixel in a display is dithered to the next adjacent unused space existing between the pixels in the display.

For viewer comfort, a strap adjustment can be provided, allowing both a one-time fastener positioning adjustment and a flexible stretch band.

The invention has been described in detail with particular reference to a presently preferred embodiment, but it will be understood that variations and modifications can be effected within the spirit and scope of the disclosure. The presently disclosed embodiments are therefore considered in all respects to be illustrative and not restrictive. The scope of the invention is indicated by any appended claims, and all changes that come within the meaning and range of equivalents thereof are intended to be embraced therein.

What is claimed is:

1. A wearable display apparatus comprising:

a catadioptric pupil-forming optical system including:

an image generator that forms a 2D image;

a partially transmissive mirror having a curved focal surface and a partially transmissive curved reflective surface;

a beam splitter disposed to reflect light toward the partially transmissive mirror, wherein the curved focal surface is defined in a space between the partially transmissive curved reflective surface and the beam splitter; and

an optical image relay positioned between the image generator and the beam splitter, and configured to conjugate the formed 2D image from the image generator to the curved focal surface of the partially transmissive mirror, the optical image relay includes:

a right-angle prism having a prism input surface, a prism output surface, and a folding surface extending from the prism input surface to the prism output surface, the

16

folding surface is configured to fold an optical path for light generated by the image generator towards the beam splitter;

a first plano-aspheric lens cemented to the prism input surface and configured to guide light from the image generator toward the folding surface and to form an aperture stop for the optical image relay at the folding surface;

a doublet positioned between the first plano-aspheric lens and the image generator;

a meniscus singlet lens positioned between the doublet and the image generator;

a concave-plano field lens positioned between the meniscus singlet and the image generator; and

a second plano-aspheric lens cemented to the prism output surface and configured to direct the light to the curved focal surface;

wherein the optical image relay, the partially transmissive mirror, and the beam splitter are configured to form an exit pupil for viewing the 2D image superimposed on a portion of a visible object scene.

2. The wearable display apparatus of claim 1, wherein the image generator includes a flat-panel display.

3. The wearable display apparatus of claim 1, wherein the image generator includes an organic light-emitting device (OLED).

4. The wearable display apparatus of claim 1, wherein the image generator includes a micro-LED array and lenslets.

5. The wearable display apparatus of claim 1, wherein the partially transmissive mirror has a conic surface shape.

6. The wearable display apparatus of claim 1, wherein the partially transmissive mirror is between about 30% to 70% transmissive.

7. The wearable display apparatus of claim 1, wherein the folding surface of the right-angle prism is coated to enhance reflection.

8. The wearable display apparatus of claim 1, wherein the beam splitter is a wire grid polarization beam splitter.

9. The wearable display apparatus of claim 1, further comprising a cylindrically curved quarter-wave plate (QWP) between the partially transmissive mirror and the beam splitter.

10. The wearable display apparatus of claim 1, wherein the catadioptric pupil-forming optical system forms a 10 mm exit pupil at a viewer's eye-box.

11. An augmented reality head-mounted-display (AR-HMD) system, comprising:

a headset including an optical module housing configured to be disposed above an eyes of a viewer;

a left-eye catadioptric pupil-forming optical system coupled to the optical module housing and configured to display a first image to a left-eye of the viewer; and

a right-eye catadioptric pupil-forming optical system coupled to the optical module housing and configured to display a second image to a right-eye of the viewer; wherein each left-eye and right-eye catadioptric pupil-forming optical system includes:

an image generator that forms a 2D image;

a partially transmissive mirror having a curved focal surface and a partially transmissive curved reflective surface;

a beam splitter disposed to reflect light toward the partially transmissive mirror, wherein the curved focal surface is defined in a space between the partially transmissive curved reflective surface and the beam splitter; and

17

an optical image relay positioned between the image generator and the beam splitter, and configured to conjugate the formed 2D image from the image generator to the curved focal surface of the partially transmissive mirror; the optical image relay includes:

- a right-angle prism having a prism input surface, a prism output surface, and a folding surface extending from the prism input surface to the prism output surface, the folding surface is configured to fold an optical path for light generated by the image generator towards the beam splitter;
- a first plano-aspheric lens cemented to the prism input surface and configured to guide light from the image generator toward the folding surface and to form an aperture stop for the optical image relay at the folding surface;
- a doublet positioned between the first plano-aspheric lens and the image generator;
- a meniscus singlet lens positioned between the doublet and the image generator;
- a concave-plano field lens positioned between the meniscus singlet and the image generator; and
- a second plano-aspheric lens cemented to the prism output surface and configured to direct the light to the curved focal surface;

wherein the optical image relay, the partially transmissive mirror, and the beam splitter are configured to form an

18

exit pupil for viewing the 2D image superimposed on a portion of a visible object scene.

12. The AR-HMD system of claim 11, wherein the image generator includes a flat-panel display.

13. The AR-HMD system of claim 11, wherein the image generator includes an organic light-emitting device (OLED).

14. The AR-HMD system of claim 11, wherein the image generator includes a micro-LED array and lenslets.

15. The AR-HMD system of claim 11, wherein the partially transmissive mirror has a conic surface shape.

16. The AR-HMD system of claim 11, wherein the partially transmissive mirror is between about 30% to 70% transmissive.

17. The AR-HMD system of claim 11, wherein the folding surface of the right-angle prism is coated to enhance reflection.

18. The AR-HMD system of claim 11, wherein the beam splitter is a wire grid polarization beam splitter.

19. The AR-HMD system of claim 11, wherein each left-eye and right-eye catadioptric pupil-forming optical system includes a cylindrically curved quarter-wave plate (QWP) between the partially transmissive mirror and the beam splitter.

20. The AR-HMD system of claim 11, wherein each left-eye and right-eye catadioptric pupil-forming optical system forms a 10 mm exit pupil at a viewer's eye-box.

* * * * *